# TM Forum Specification

# Customer360 User Guide

## TMF717

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# Introduction

The following document is the user guide of the REST API for Customer360. It includes the model definition as well as the available operations.

## API Overview and Objectives

The Customer360 API provides a comprehensive, unified view of customer information by aggregating data from multiple sources into a single, read-only resource. This API serves the following key objectives:

- **Holistic Customer View**: Combines customer information from various domains (accounts, products, bills, tickets, appointments, agreements, etc.) into a single consolidated resource
- **Operational Efficiency**: Enables applications to retrieve complete customer context in a single API call, reducing the need for multiple API requests across different systems
- **Enhanced Customer Experience**: Supports customer-facing applications (such as call centers, self-service portals, and mobile apps) with comprehensive customer data for better service delivery
- **Read-Only Access**: Provides a safe, non-intrusive way to access aggregated customer data without modifying underlying systems

## Main Concepts and Resources

The Customer360 API manages the following main resource:

- **Customer360**
  - Customer360 is a read-only resource that aggregates comprehensive information about a customer from multiple domains
  - The unique identifier of the Customer360 resource equals the customer identifier in the Customer Management API
  - Main components include:
    - Customer basic information (name, status, contact details, related parties)
    - Accounts (billing accounts, settlement accounts)
    - Agreements (commercial agreements, service level agreements)
    - Appointments (scheduled interventions, service visits)
    - Customer bills (invoices, payment status)

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- Loyalty accounts (loyalty points, rewards balance)
- Party interactions (customer service interactions, store visits)
- Payment methods (credit cards, bank accounts)
- Products (subscribed products and services)
- Product orders (order history, order status)
- Usage records (product usage and consumption patterns)
- Quotes (customer quotes, pricing information)
- Service problems (service issues, problem tracking)
- Trouble tickets (open issues, service requests)

**Important Note on Value Objects:** All Value Object (VO) entities defined within the Customer360 domain represent subsets of their corresponding source entities. Consequently, the attributes declared within each VO schema object constitute a subset of the attribute definitions present in their respective originating schema objects.

The Customer360 API performs the following operations:

- Retrieval of complete customer 360-degree view
- List/query operations with filtering capabilities
- Event notifications for customer data changes

## API Dependencies

This API aggregates information from multiple TM Forum APIs and assumes that detailed information can be obtained by accessing the following APIs:

- **TMF629 Customer Management API**: For detailed customer information
- **TMF666 Account Management API**: For account details and financial information
- **TMF651 Agreement Management API**: For agreement and contract information
- **TMF646 Appointment API**: For appointment scheduling information
- **TMF678 Customer Bill Management API**: For billing and invoice details
- **TMF658 Loyalty Management API**: For loyalty account and rewards information
- **TMF683 Party Interaction API**: For customer interaction history
- **TMF670 Payment Method API**: For payment method information
- **TMF637 Product Inventory API**: For product and service subscription details

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- **TMF622 Product Ordering Management API**: For product order information
- **TMF635 Usage Management API**: For product usage and consumption information
- **TMF648 Quote Management API**: For customer quote and pricing information
- **TMF656 Service Problem Management API**: For service problem tracking information
- **TMF621 Trouble Ticket API**: For issue tracking and service request information

## Relationship with Other TM Forum Assets

The Customer360 API is part of the TM Forum Open API suite and aligns with:

- **TM Forum Information Framework (SID)**: The data model follows SID principles for customer, account, product, and service entities
- **TM Forum Open API Guidelines (TMF630)**: Implements standard REST patterns, polymorphism support, and extension mechanisms
- **TM Forum Business Process Framework (eTOM)**: Supports Customer Relationship Management (CRM) processes in the Operations and Fulfillment domains

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# Sample Use Cases

The Customer360 API provides a unified view of customer information across multiple domains, enabling various business scenarios. This section describes detailed use cases that demonstrate how the API can be utilized in real-world applications.

## Use Case 1: Call Center Customer Service Representative Dashboard

### Business Context

In a telecommunications service provider environment, call center representatives need immediate access to comprehensive customer information to provide efficient and personalized service. When a customer calls with a question or issue, the representative must quickly understand the customer's complete context including their products, billing status, open tickets, scheduled appointments, and interaction history.

### Actors

- Call Center Representative (primary actor)
- Customer (initiating actor)
- Customer360 API (supporting system)
- Call Center Application (supporting system)

### Preconditions

- Customer is identified and authenticated (customer ID: 1140)
- Call center representative has access to the customer service application
- Customer360 data has been aggregated from source systems

### Main Flow

1. Customer "Moon Football Club" (ID: 1140) calls the call center regarding their internet service
2. Call center representative enters or receives the customer identifier (1140) in the system
3. The call center application invokes: `GET /customer360/1140`
4. The Customer360 API returns a comprehensive customer profile containing:
    - **Customer Information**: Name "Moon Football Club", status "Approved", contact

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details including phone (01 09 75 83 51) and address (15 Rue des Canards, Paris)

- **Accounts**: Business account (ID: 5430) with name "Home Account" in "Inactive" state
- **Products**: Multiple subscribed products including:
  - Internet service (ID: 1111) with status "Active"
  - Mobile service (ID: 2222) with status "Suspended"
- **Customer Bills**: Recent bills including:
  - Bill CB-123 with status "sent" (monthly category)
  - Bill CB-123 with status "settled" (normal category)
- **Trouble Tickets**: Open ticket (ID: 7654) with description "Internet connection issue" and status "In Progress"
- **Appointments**: Scheduled intervention (ID: 21) for "Fix an internet connexion problem" on 2018-02-15 from 14:00 to 16:00, status "confirmed"
- **Agreements**: Active agreement "Summer Contract Agreement" (ID: 28) valid until 2028-04-25
- **Loyalty Account**: Active loyalty account (ID: 12345) with balance of 50 EUR
- **Payment Methods**: Preferred gold mastercard (ID: 12345) with validation code
- **Party Interactions**: Recent store visit (ID: 123) on 2019-10-02 with reason "The user wanted to buy TV and internet plan"

5. The representative reviews the consolidated information and identifies:
   - An open trouble ticket about internet connectivity
   - A scheduled appointment for the same day to fix the issue
   - A suspended mobile service that might be related to billing
   - An inactive account status that requires attention
6. The representative provides informed assistance to the customer, referencing:
   - The scheduled technician visit later that day
   - The account status and suggests reactivation
   - The loyalty points available for service credits

**Alternative Flows**

- **Partial Data Available**: If some data sources are unavailable, the API returns available information with notifications about missing data domains

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- **Customer Not Found**: If customer ID doesn't exist, API returns 404 error
- **Multiple Customers**: Representative can use query parameters to list and filter customers: `GET /customer360?customer.name=Moon&customer.status=Approved`

**Postconditions**

- Representative has complete customer context
- Customer receives informed and personalized service
- Call handling time is reduced due to single API call
- Customer satisfaction is improved

**Related API Examples**

This use case is illustrated by the following examples in the API documentation:

- **Resource Example**: `Customer360.example.json` - Shows the complete Customer360 resource structure for customer ID 1140
- **Retrieve Operation**: `Customer360_retrieve_1_response.sample.json` - Demonstrates the GET operation response for retrieving a single customer's 360-degree view
- **List Operation**: `Customer360_list_1_response.sample.json` - Shows how to query and filter multiple customer records
- **Create Event Notification**: `customer360_create_event_example.json` - Illustrates the notification sent when a new Customer360 resource is created
- **Attribute Value Change Event**: `customer360_AttributeValueChange_event_example.json` - Shows notification when customer data is updated
- **Delete Event Notification**: `customer360_delete_event_example.json` - Demonstrates the notification when a Customer360 resource is removed

## Use Case 2: Self-Service Customer Portal

**Business Context**

Customers access a self-service web portal to view their complete profile, manage their services, and track their interactions with the service provider. The portal uses the Customer360 API to present a unified dashboard.

**Main Flow**

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1. Customer logs into the self-service portal
2. Portal application retrieves Customer360 data for the authenticated customer
3. Dashboard displays:
   - Account summary with current status and balance
   - Active products and services
   - Recent bills and payment history
   - Loyalty points and rewards
   - Scheduled appointments
   - Open tickets and their status
4. Customer can navigate to detailed management APIs for specific actions (e.g., pay bill, report issue)

**Benefits**

- Single API call reduces page load time
- Consistent customer experience across channels
- Real-time aggregated view of customer data

## Use Case 3: Proactive Customer Retention

**Business Context**

A retention team uses analytics dashboards powered by Customer360 API to identify at-risk customers and take proactive measures.

**Main Flow**

1. Retention application queries Customer360 API with filters: `GET /customer360?customer.status=Active&fields=customer,products,customerBill,troubleTicket`
2. Application analyzes patterns:
   - Multiple open trouble tickets
   - Suspended or cancelled products
   - Overdue bills
   - Reduced usage patterns
3. Retention team receives alerts for at-risk customers

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4. Team retrieves complete Customer360 view for each at-risk customer
5. Team designs personalized retention offers based on:
    - Customer's product portfolio
    - Loyalty account balance
    - Agreement terms
    - Interaction history

**Benefits**

- Early identification of churn risk
- Data-driven retention strategies
- Improved customer lifetime value

## Additional Use Cases

The Customer360 API can support many other scenarios including:

- **Mobile Application Customer View**: Mobile apps displaying customer information on smartphones
- **Partner Portal Integration**: Channel partners accessing customer information for support
- **Marketing Campaign Personalization**: Marketing systems using customer profiles for targeted campaigns
- **Billing Dispute Resolution**: Billing teams accessing complete customer context during dispute handling
- **Service Activation Workflows**: Order management systems checking customer eligibility and context

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# Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities. In the Customer360 API context, polymorphism is demonstrated through contact medium types and reference entities.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

## Meta-Attributes for Polymorphism

The @type attribute provides a way to represent the actual class type of an entity. For example, within the Customer360 resource, the contactMedium collection can contain different types of contact mediums:

- **PhoneContactMedium** - for phone contact information
- **GeographicAddressContactMedium** - for physical address information
- **EmailContactMedium** - for email contact information

All these types inherit from the base **ContactMedium** entity. The @type attribute indicates which specific type is being used. All resources and sub-resources of this API have @type attributes that can be provided when this is useful.

**Example from Customer360 API:**

In the Customer360 resource, the customer's contactMedium array demonstrates polymorphism:

```
"contactMedium": [
  {
    "@type": "PhoneContactMedium",
    "@baseType": "ContactMedium",
    "@schemaLocation": "/schema-registry/schema/PhoneContactMedium",
    "id": "2256789876",
    "preferred": true,
    "contactType": "homePhone",
    "phoneNumber": "01 09 75 83 51"
  },
  {
    "@type": "GeographicAddressContactMedium",
    "@baseType": "ContactMedium",
    "@schemaLocation": "/schema-registry/schema/GeographicAddressContactMedium",
```

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```
    "id": "12345",
    "preferred": false,
    "contactType": "homeAddress",
    "city": "Paris",
    "country": "France",
    "postCode": "75014",
    "street1": "15 Rue des Canards"
  }
]
```

The @referredType can be used within reference entities (like for instance a PartyRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as PartyRef or OrganizationRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

**Example from Customer360 API:**

In the Customer360 resource, the engagedParty reference demonstrates the use of @referredType:

```
"engagedParty": {
  "@type": "PartyRef",
  "href": "https://host:port/tmf-api/partyManagement/v4/organization/500",
  "id": "500",
  "name": "Happy Travellers",
  "@referredType": "Organization"
}
```

Here, @type indicates this is a PartyRef (the reference type), while @referredType indicates the actual referred entity is an Organization. The referred entity could also be an Individual, depending on the customer type.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

**Example from Customer360 API:**

```
{
  "@type": "Customer360",
  "@schemaLocation": "/schema-registry/schema/Customer360.yaml",
  "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
  "id": "1140"
}
```

The @baseType attribute gives a way to provide explicitly the base of class of a given

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resource that has been extended.

**Example from Customer360 API:**

In the contactMedium examples above, the @baseType attribute explicitly indicates that both PhoneContactMedium and GeographicAddressContactMedium inherit from the ContactMedium base type.

## Additional Polymorphism Examples in Customer360

The Customer360 API also demonstrates polymorphism in other areas:

- **Value Objects**: Different types of value objects (e.g., Customer360AccountVO, Customer360ProductVO, Customer360CustomerBillVO) that represent aggregated views of different domain entities
- **Related Party References**: The relatedParty collection can contain references to different party types (Organization, Individual) with appropriate @referredType values
- **Payment Methods**: Different payment method types (CreditCard, BankAccount, DigitalWallet) can be represented in the paymentMethod collection with appropriate @type discrimination

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# RESOURCE MODEL

## Managed Entity and Task Resource Models

### Customer360 resource

Object represents a consolidated Customer overview. The id of the Customer360 equals the id of the Customer.

#### Resource model

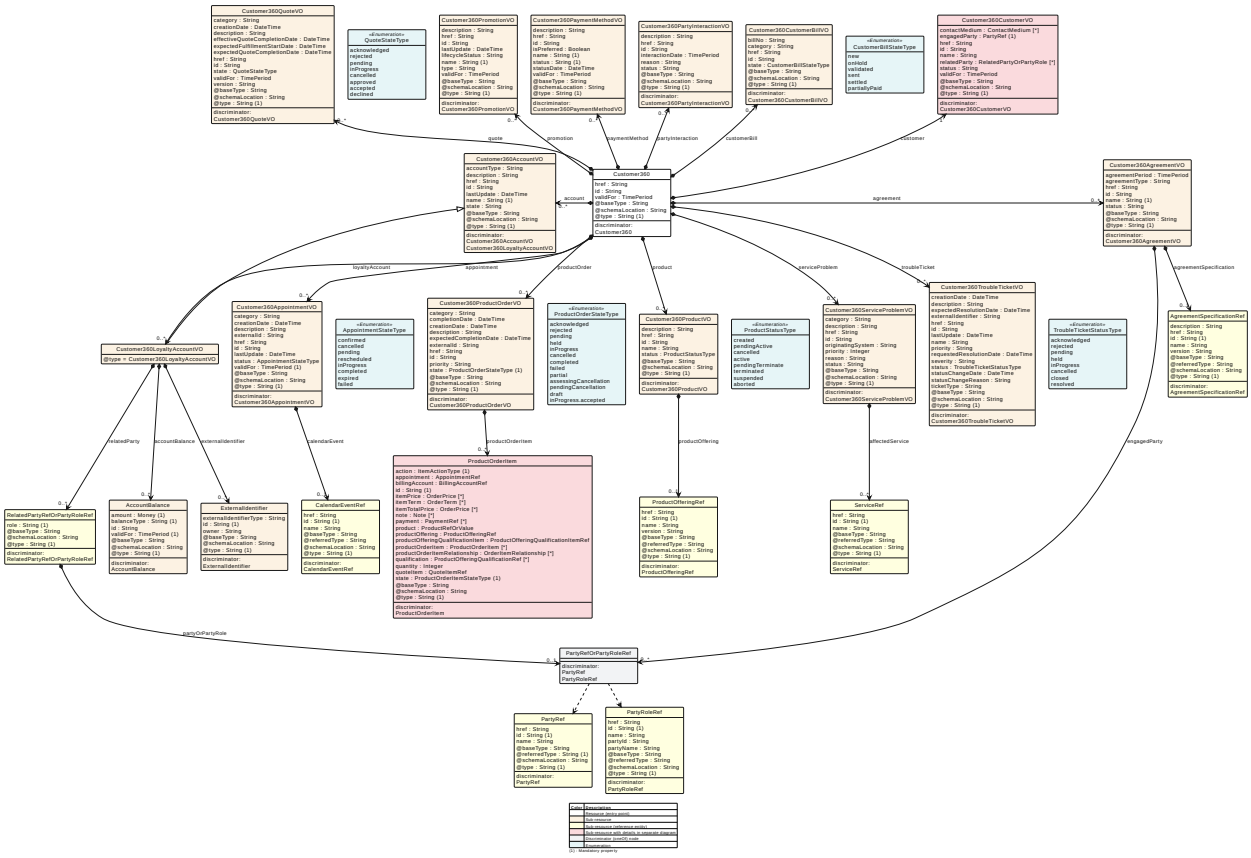


*Figure 1 - Customer360*

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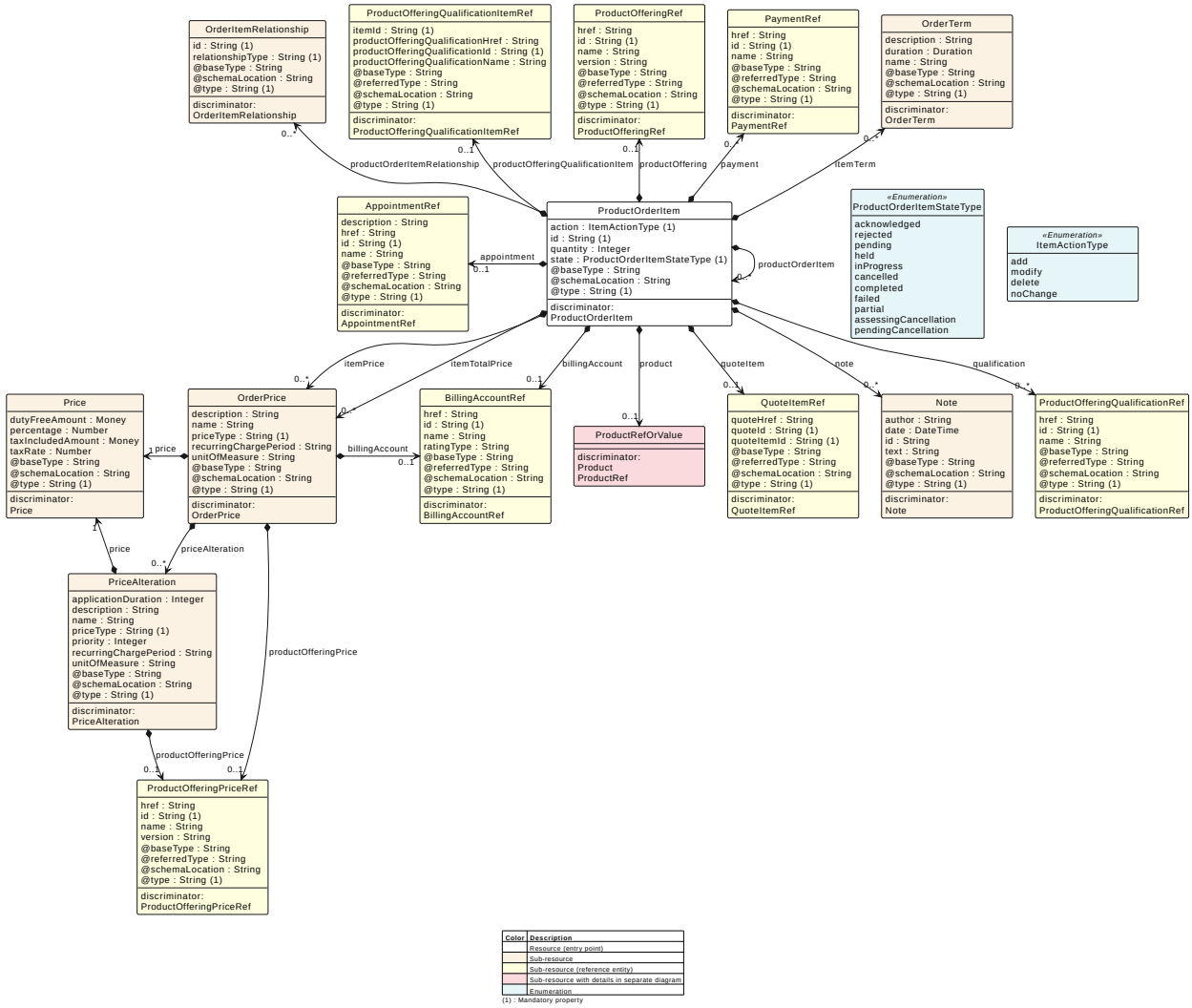



*Figure 2 - ProductOrderItem*

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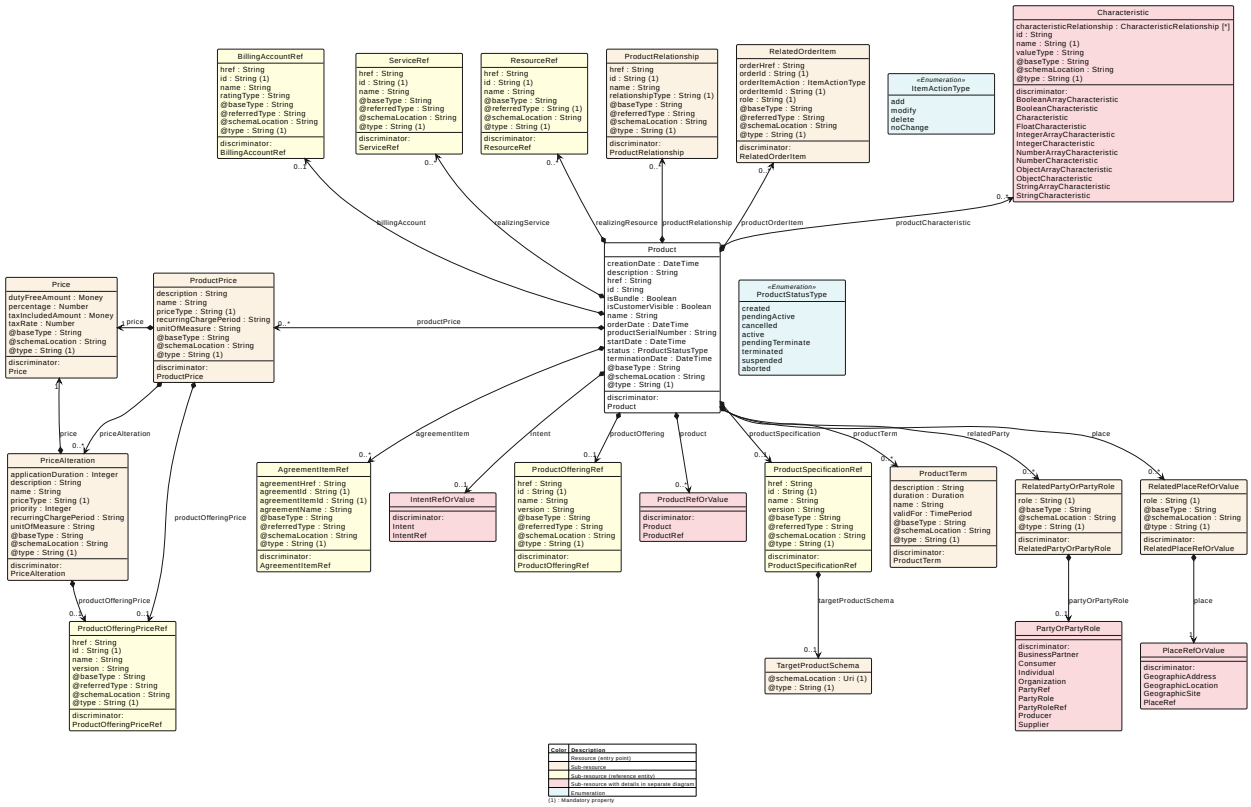



*Figure 3 - Product*

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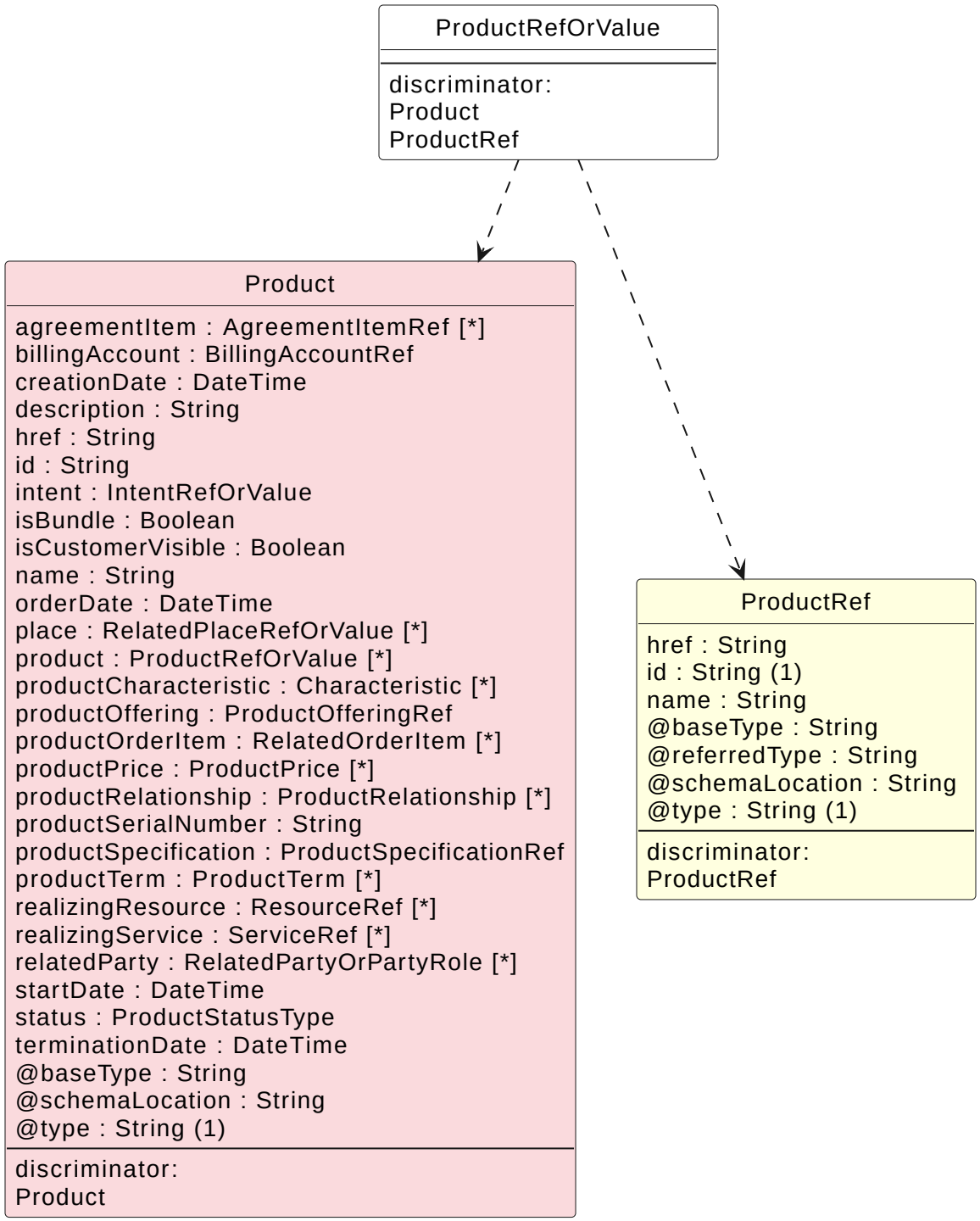



*Figure 4 - ProductRefOrValue*

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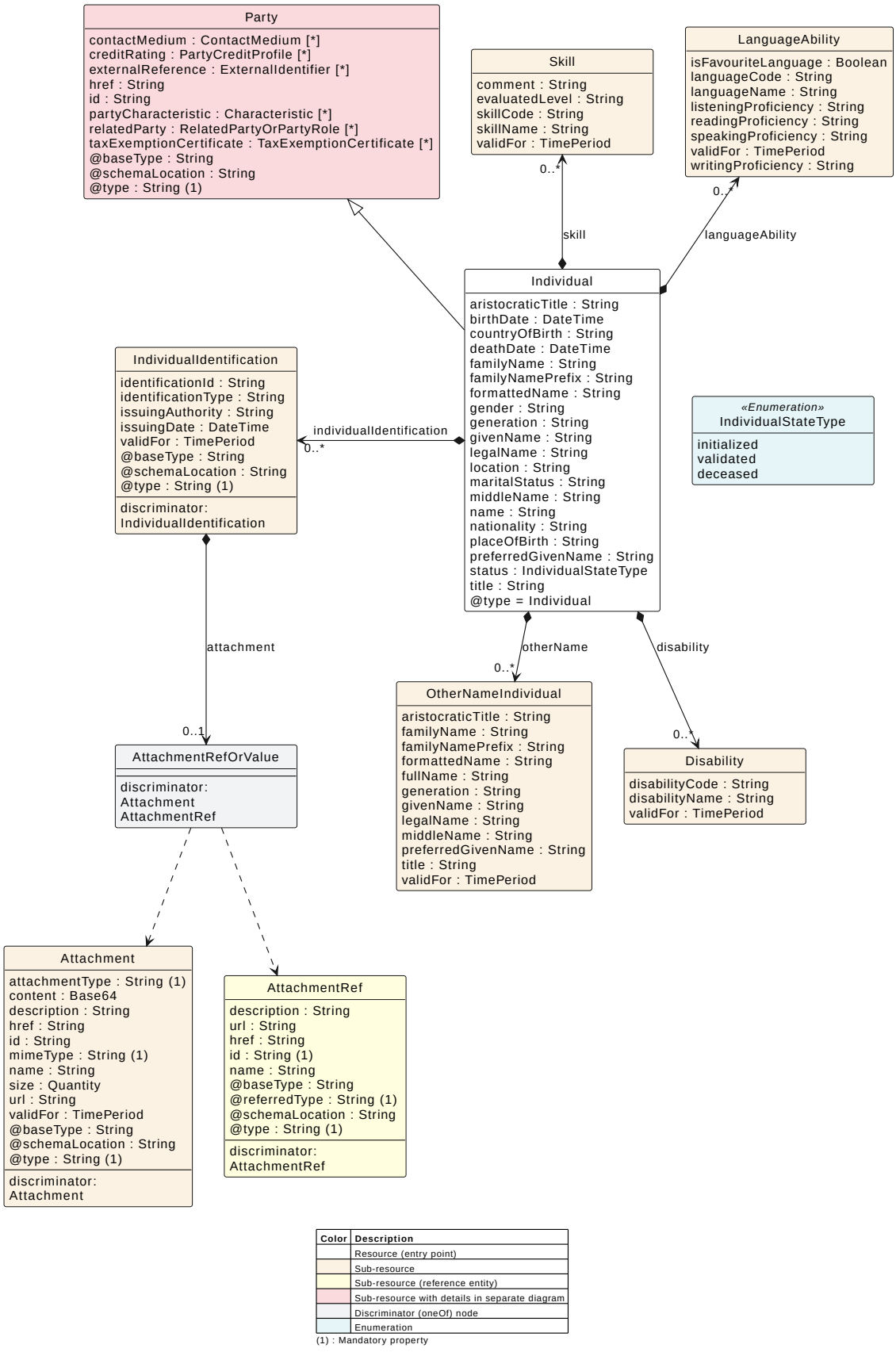



*Figure 5 - Individual*

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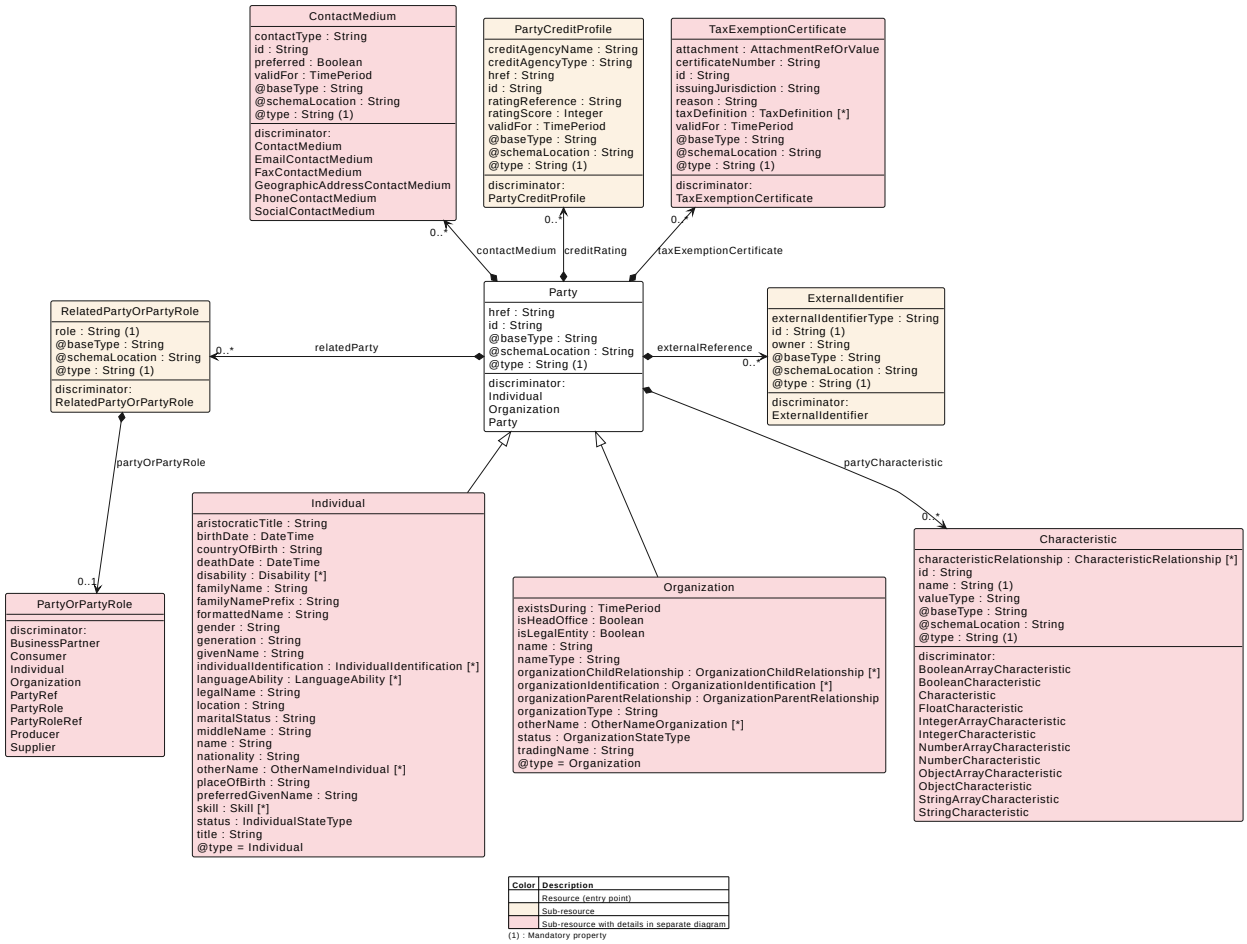



*Figure 6 - Party*

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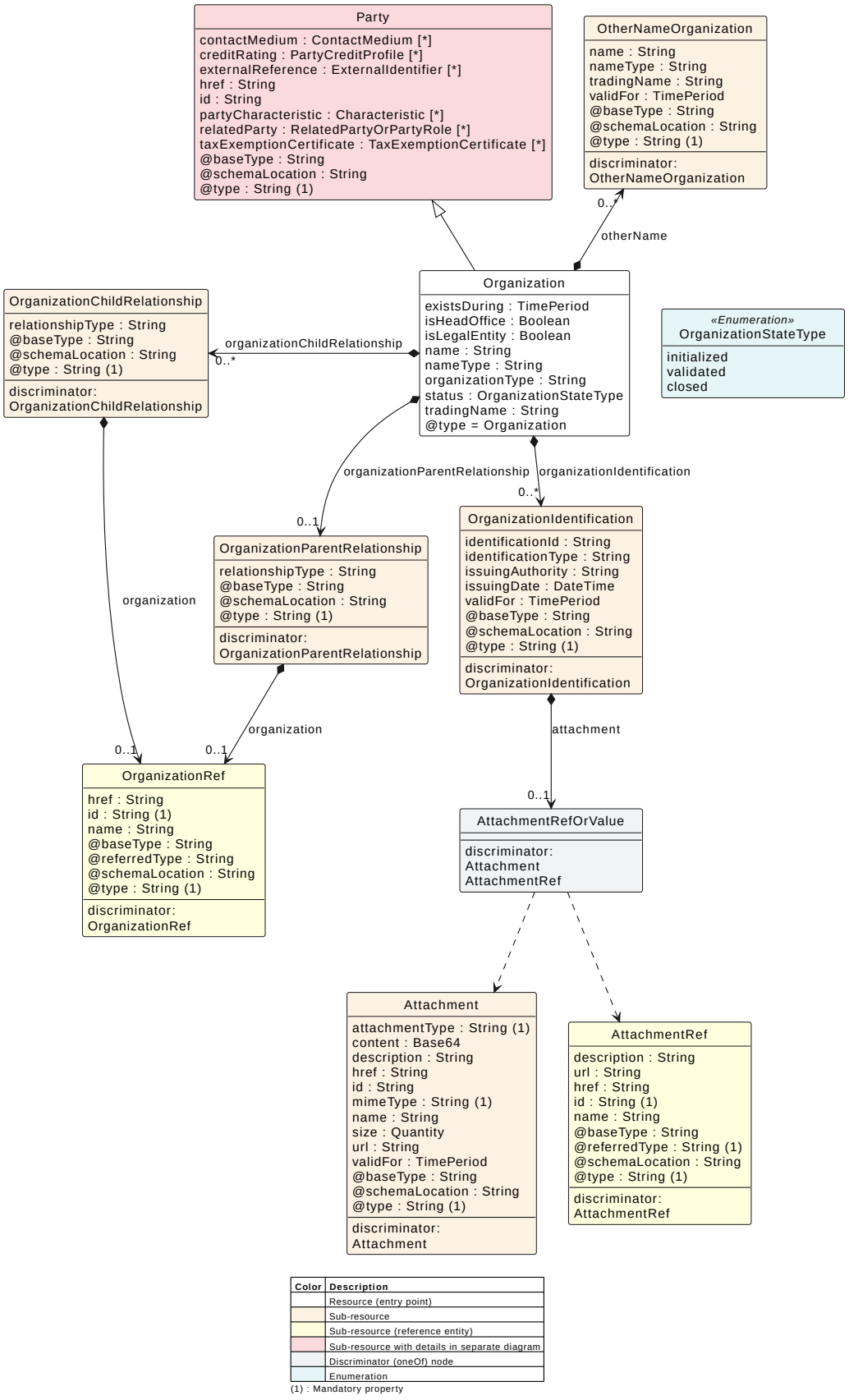



*Figure 7 - Organization*

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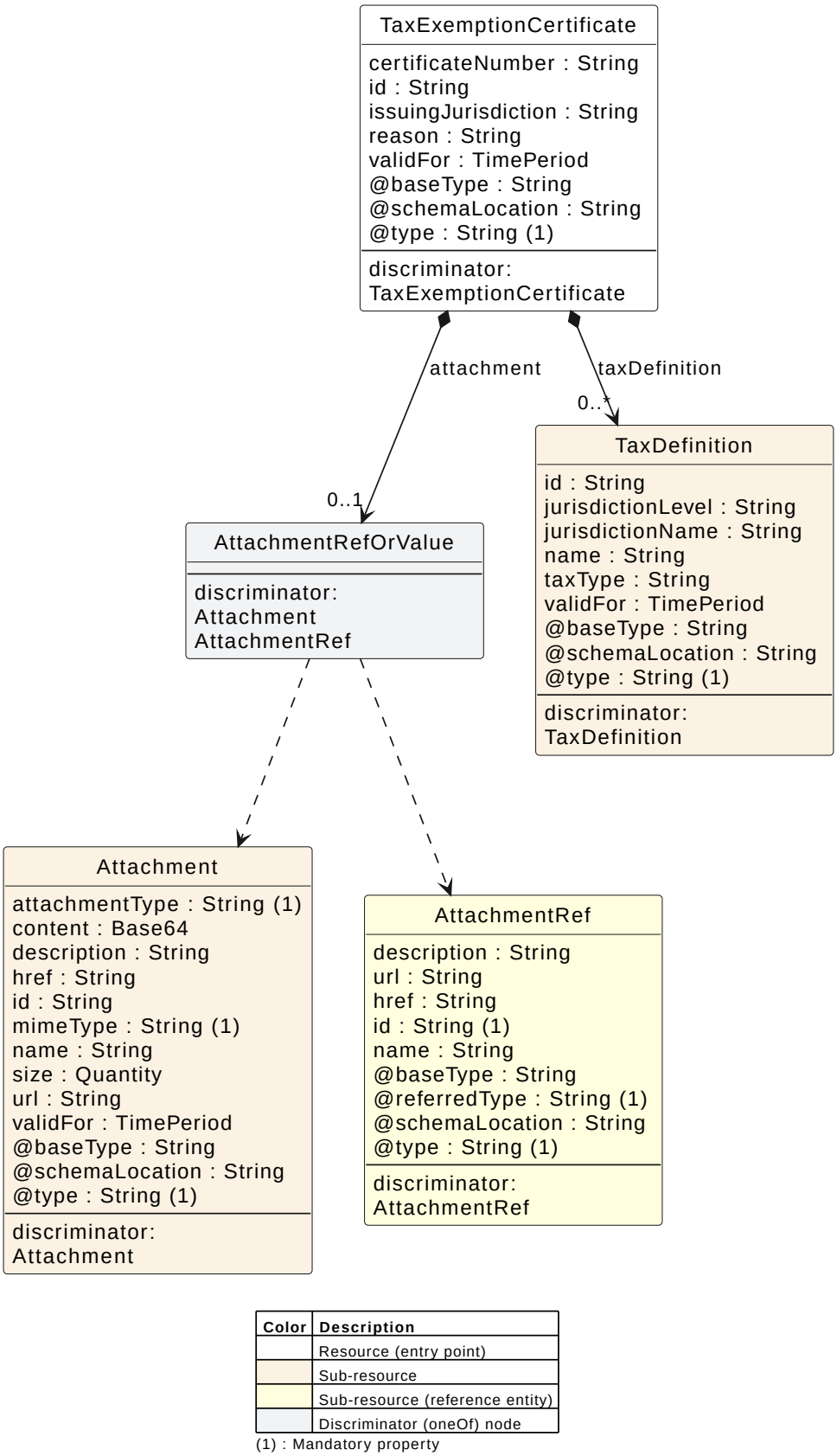



*Figure 8 - TaxExemptionCertificate*

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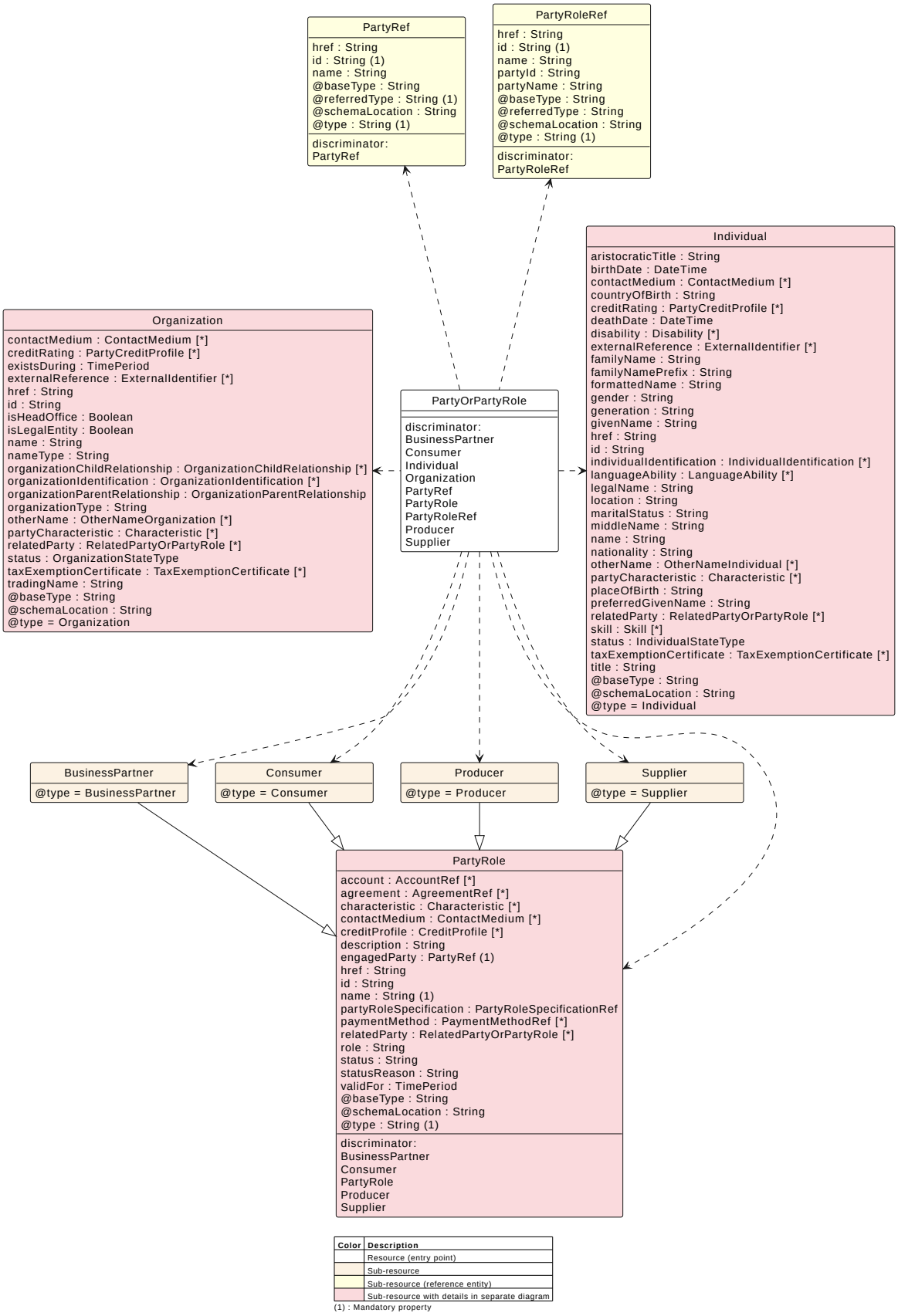



*Figure 9 - PartyOrPartyRole*

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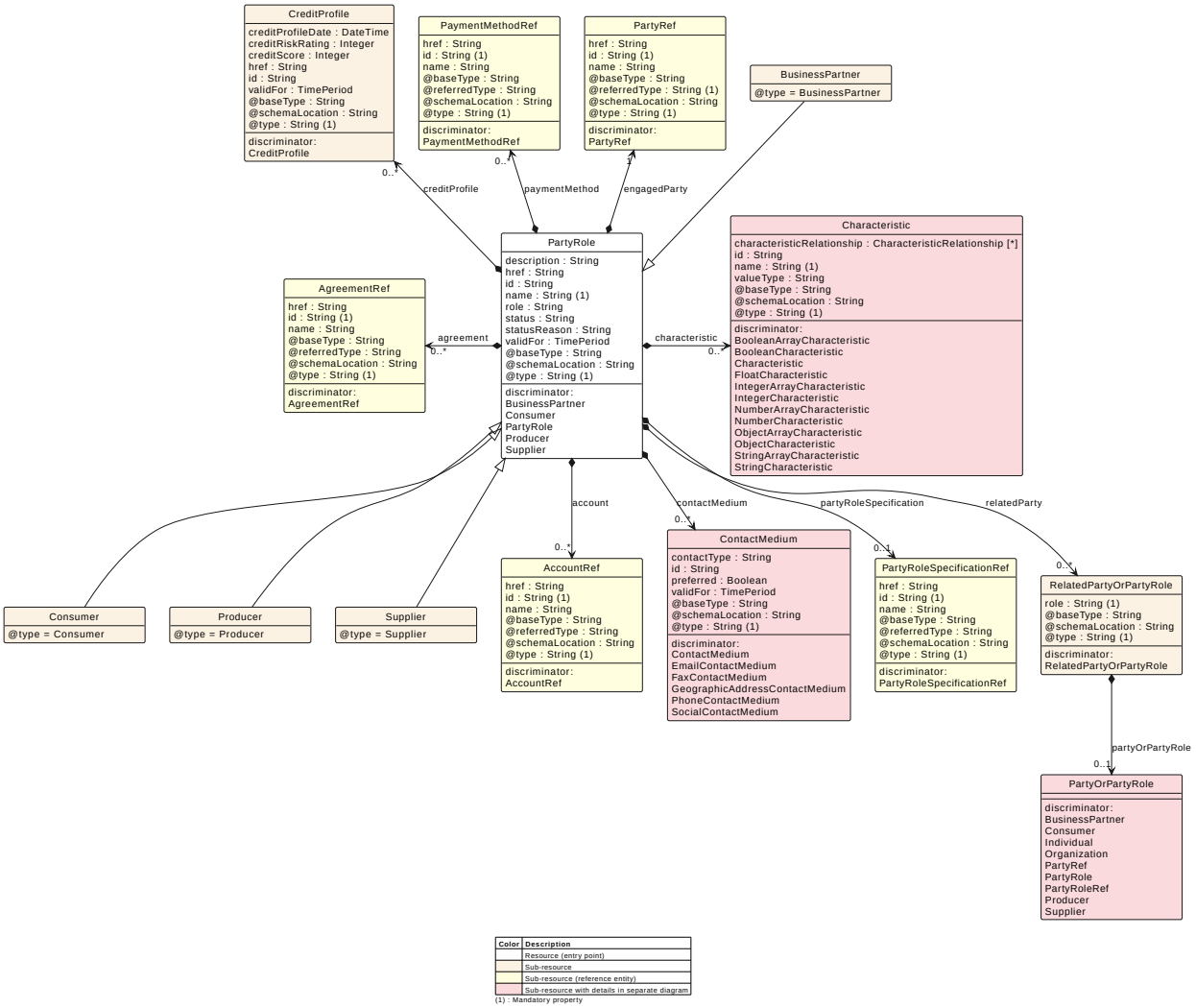



*Figure 10 - PartyRole*

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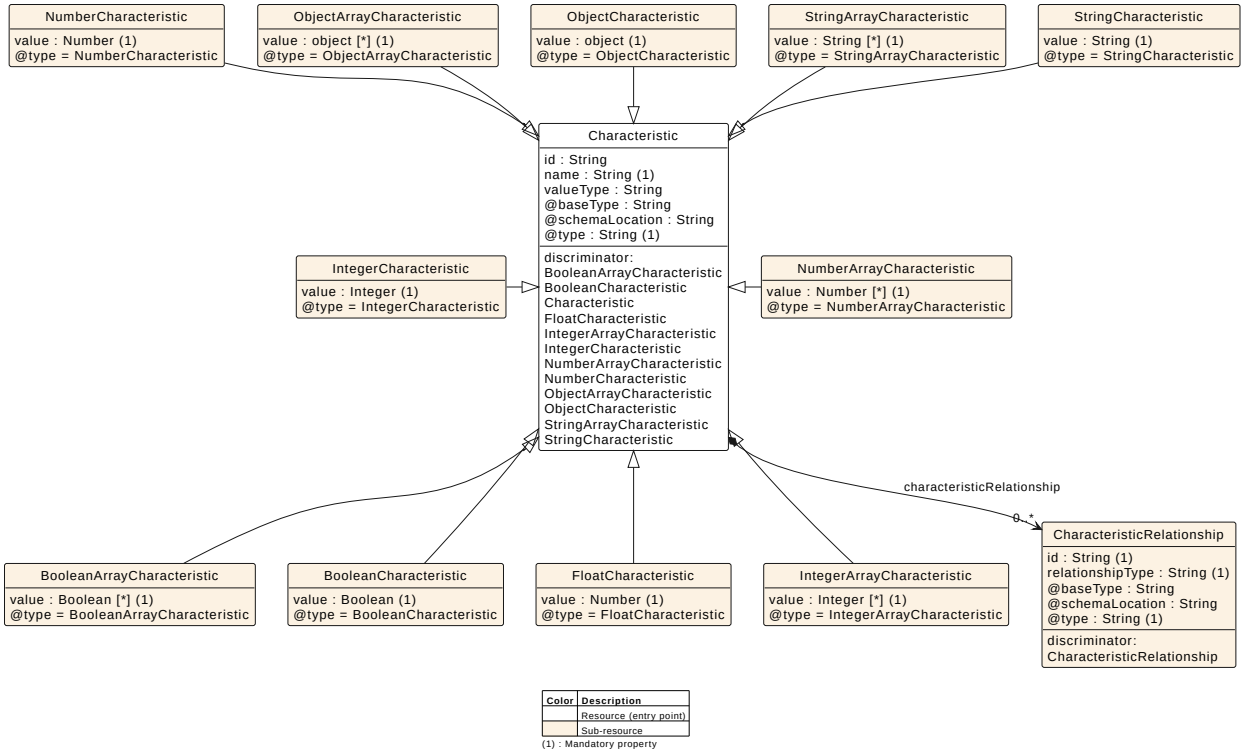



*Figure 11 - Characteristic*

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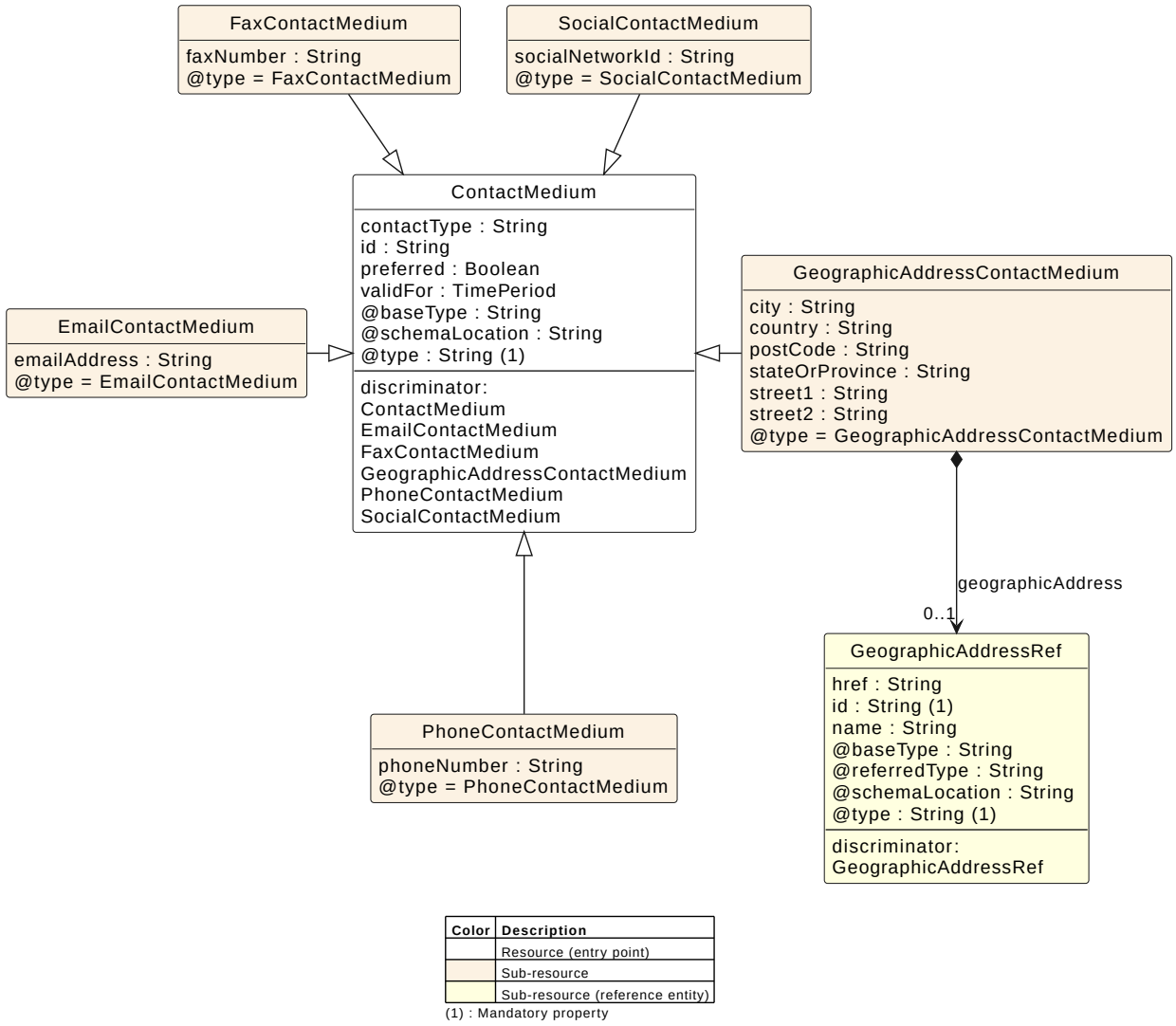



*Figure 12 - ContactMedium*

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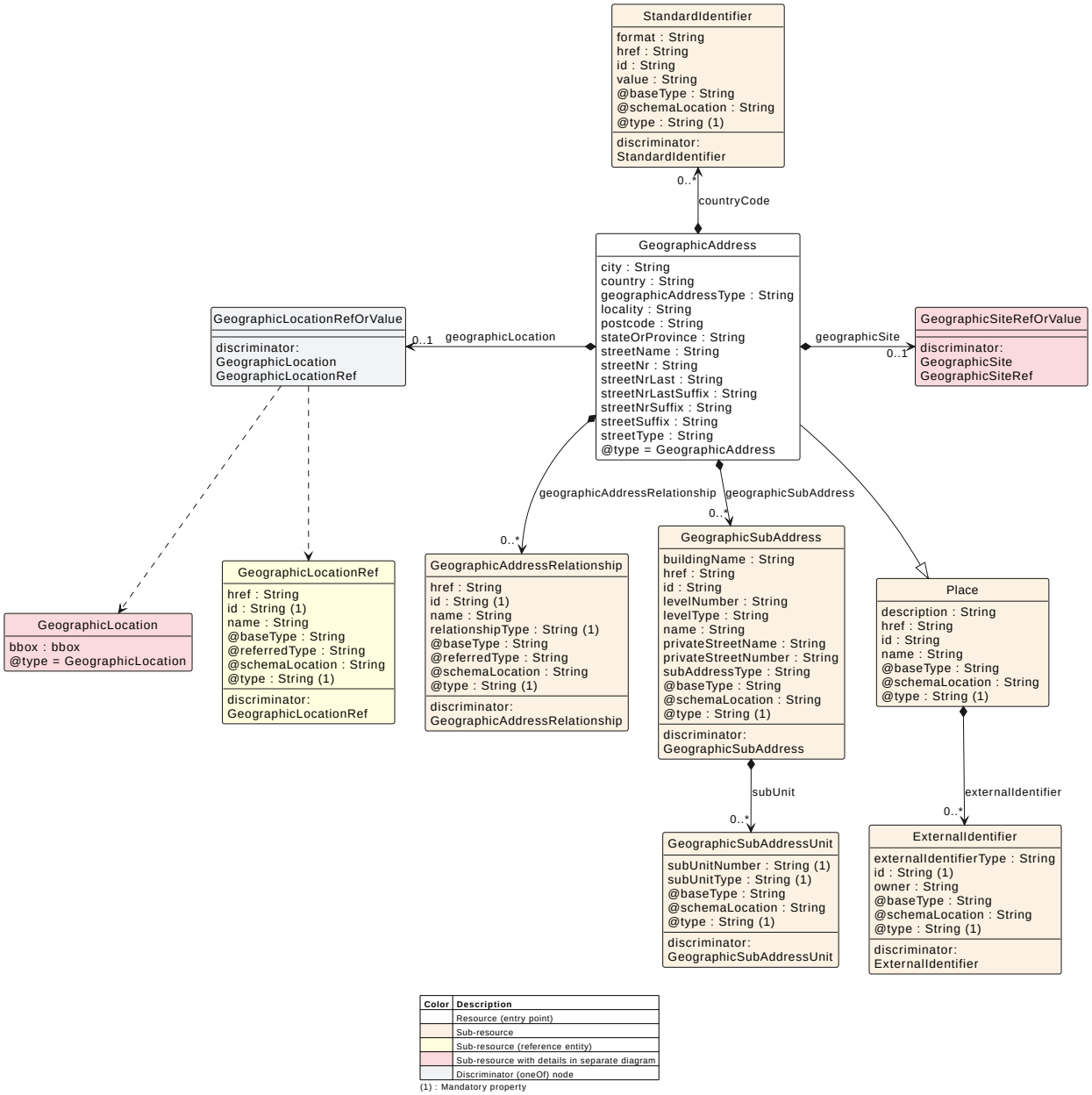



*Figure 13 - GeographicAddress*

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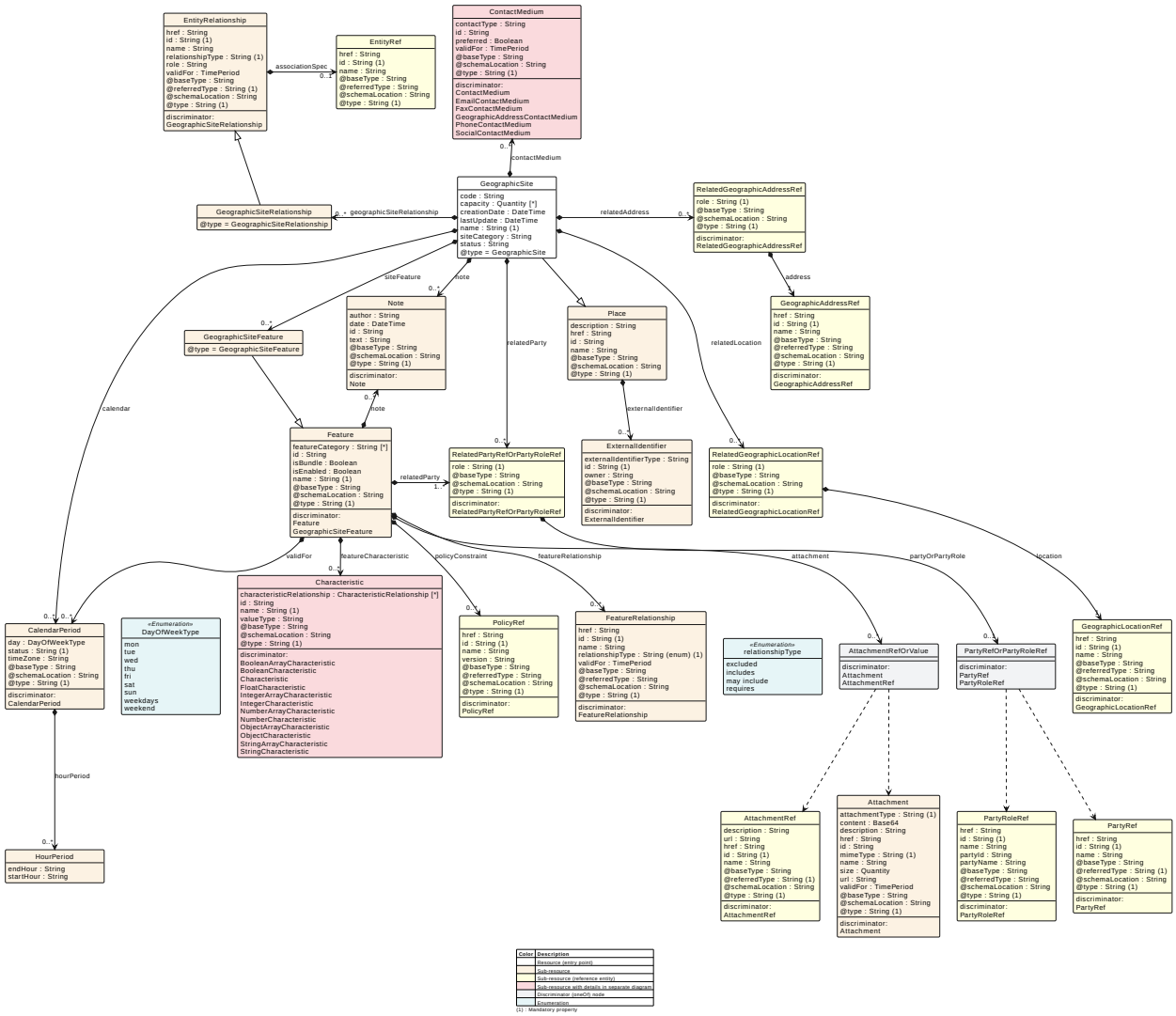



*Figure 14 - GeographicSite*

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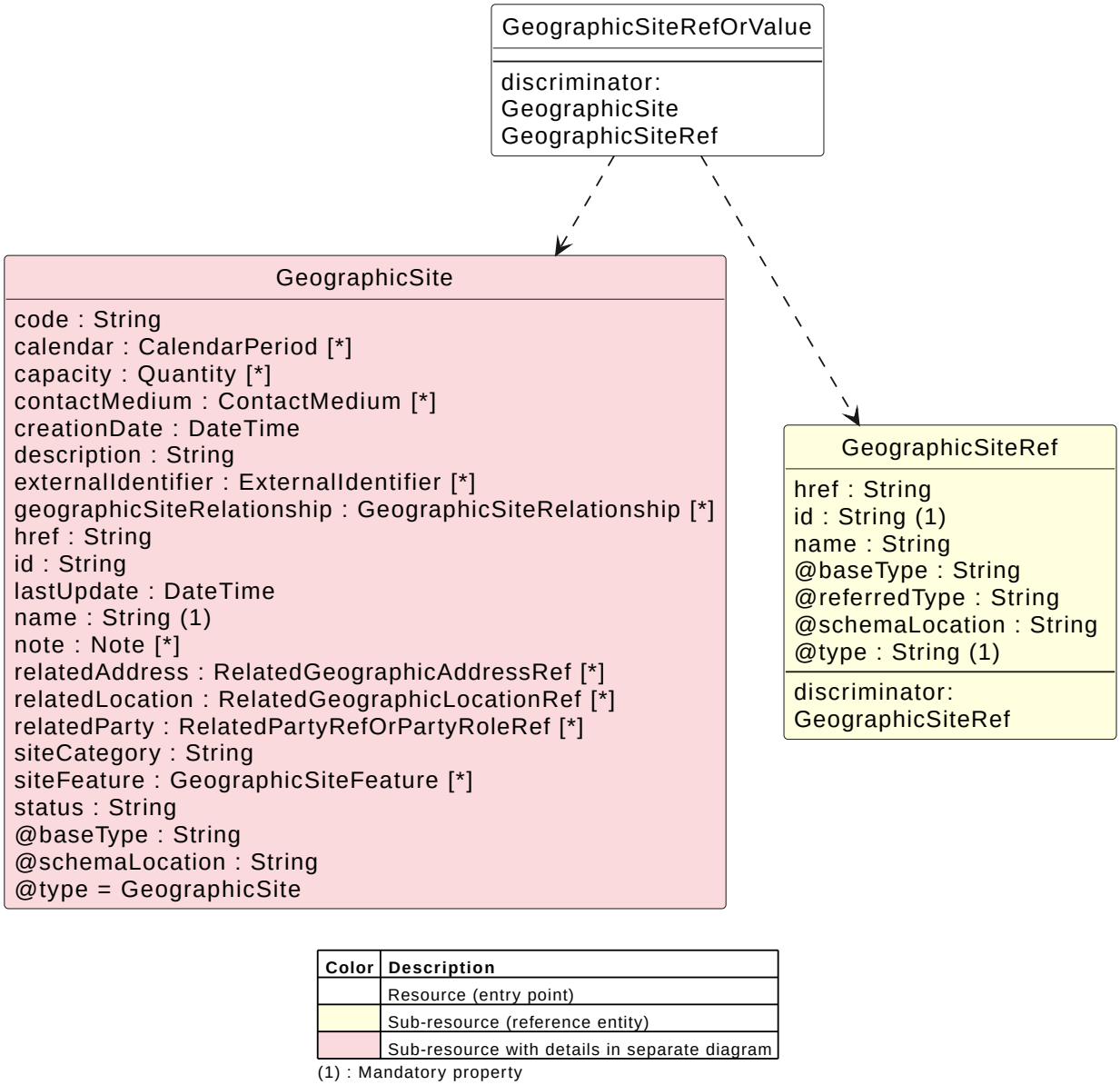



*Figure 15 - GeographicSiteRefOrValue*

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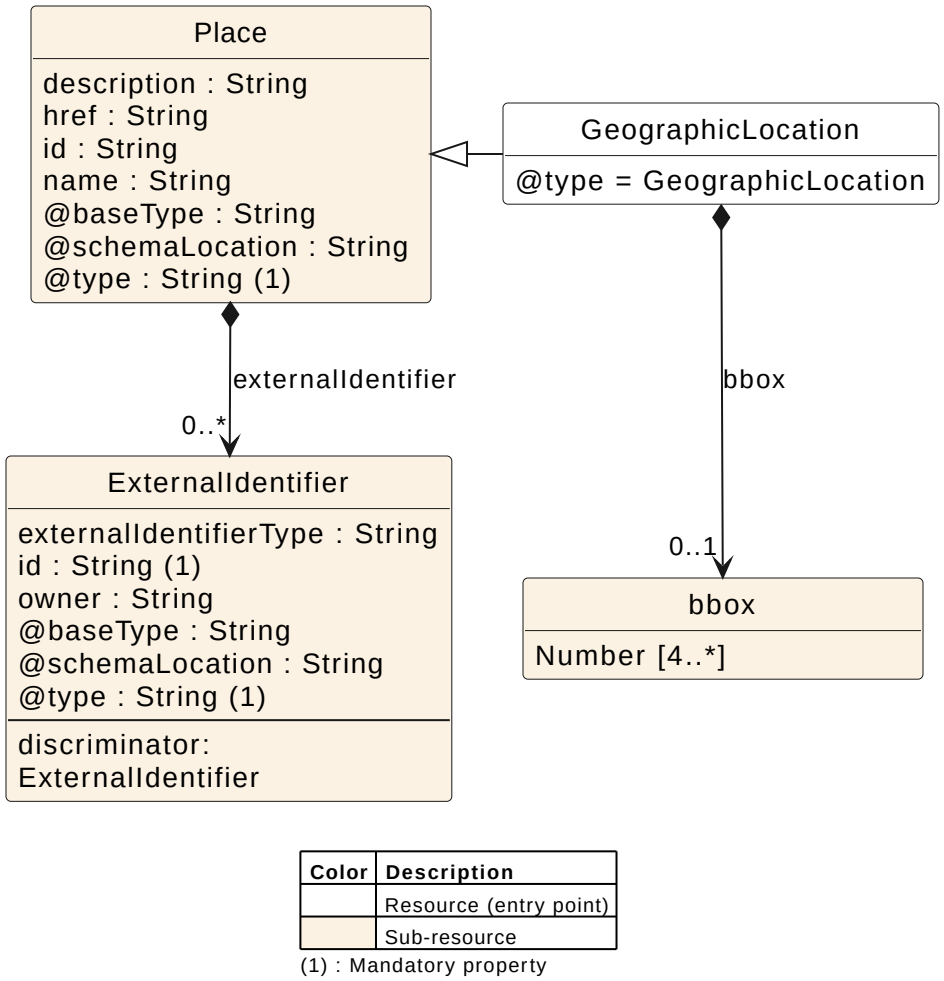



*Figure 16 - GeographicLocation*

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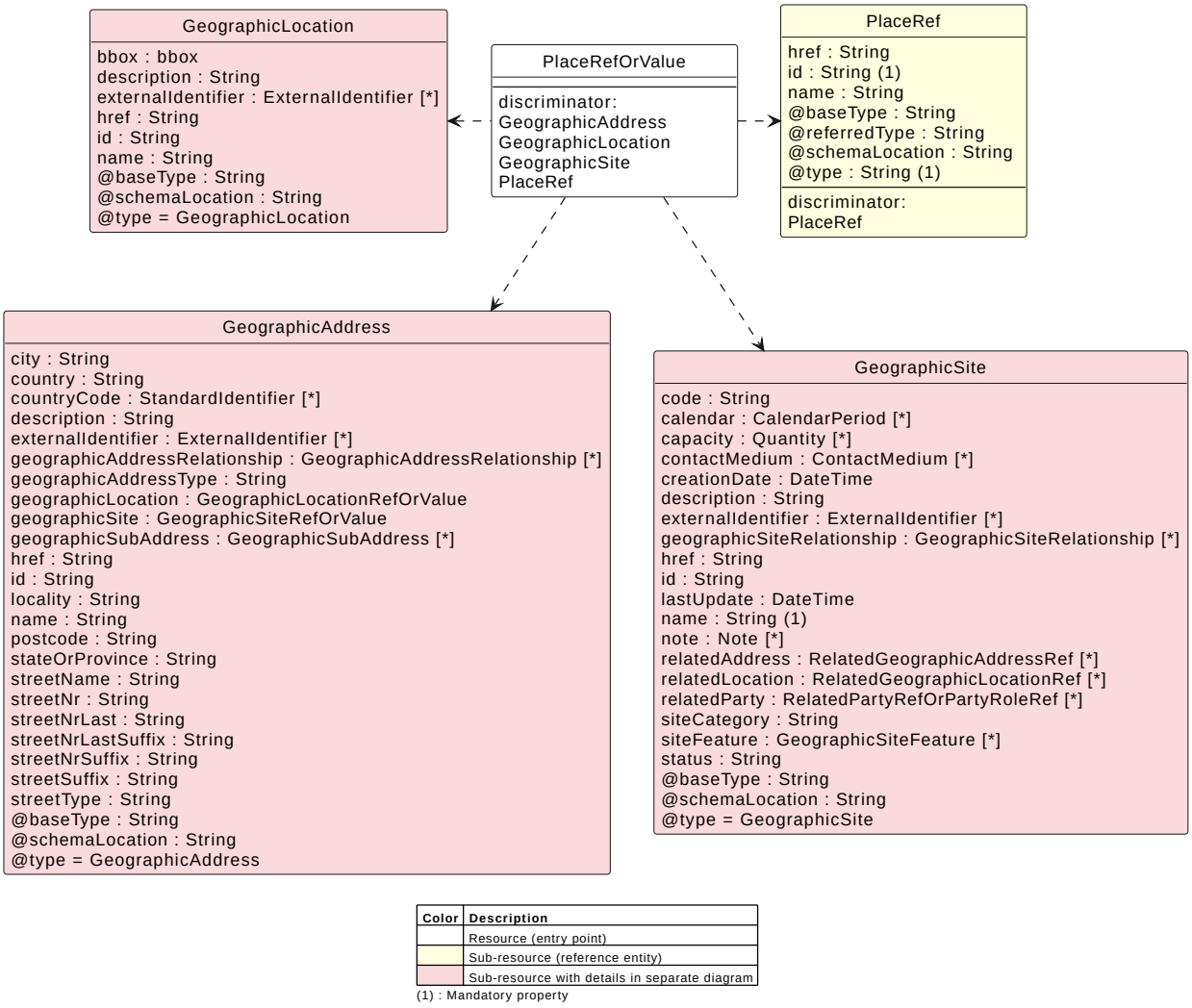



*Figure 17 - PlaceRefOrValue*

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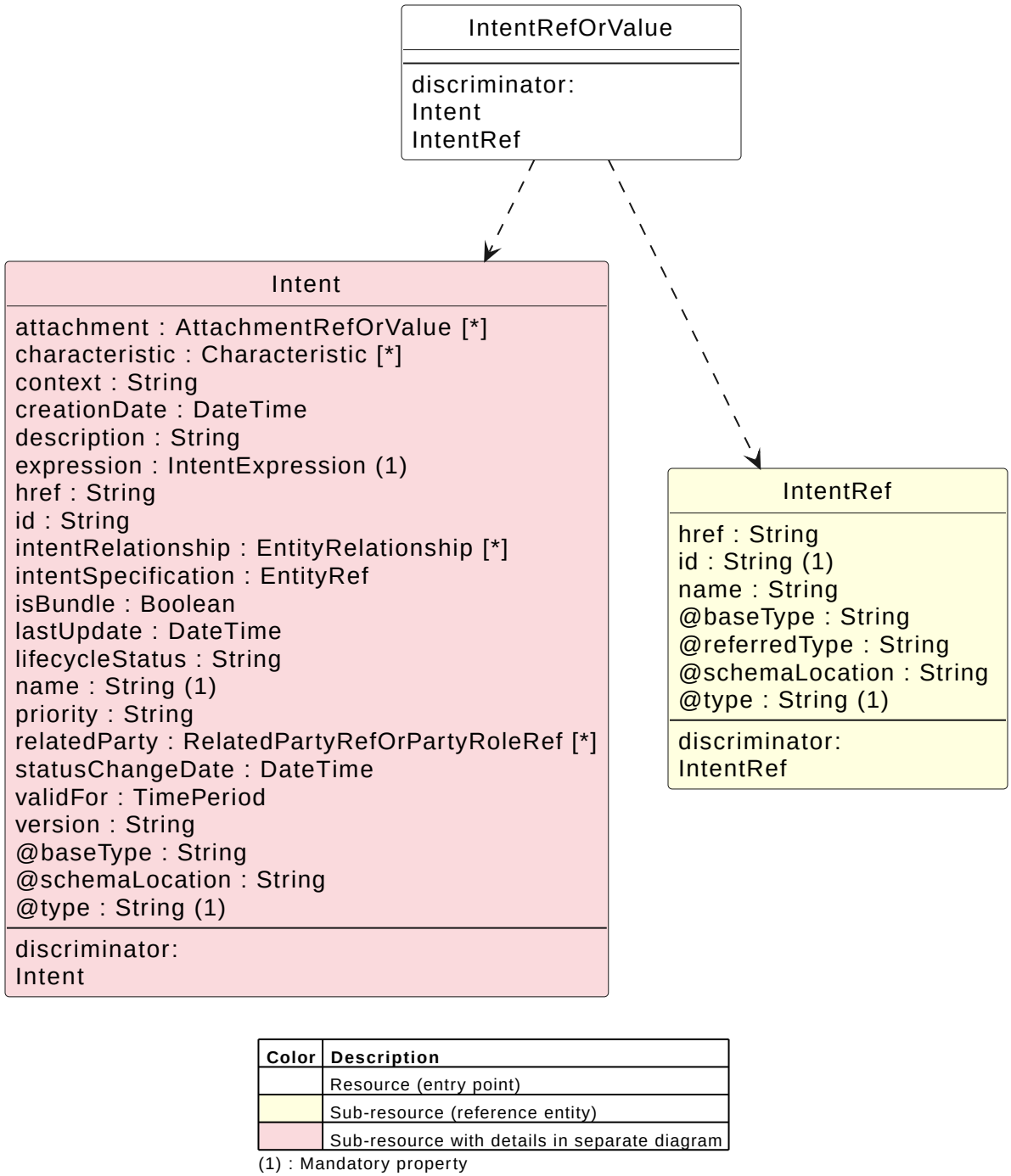



*Figure 18 - IntentRefOrValue*

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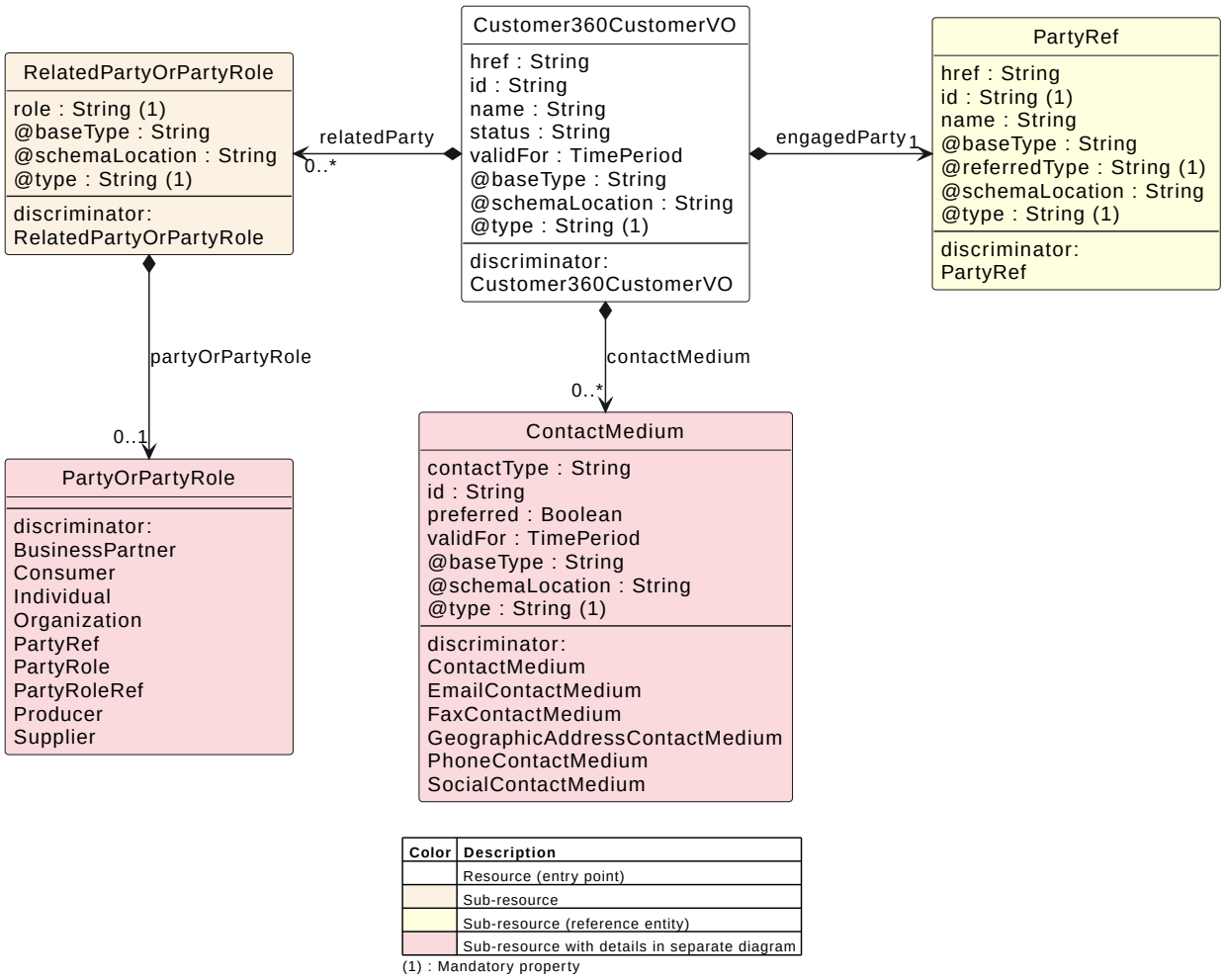



*Figure 19 - Customer360CustomerVO*

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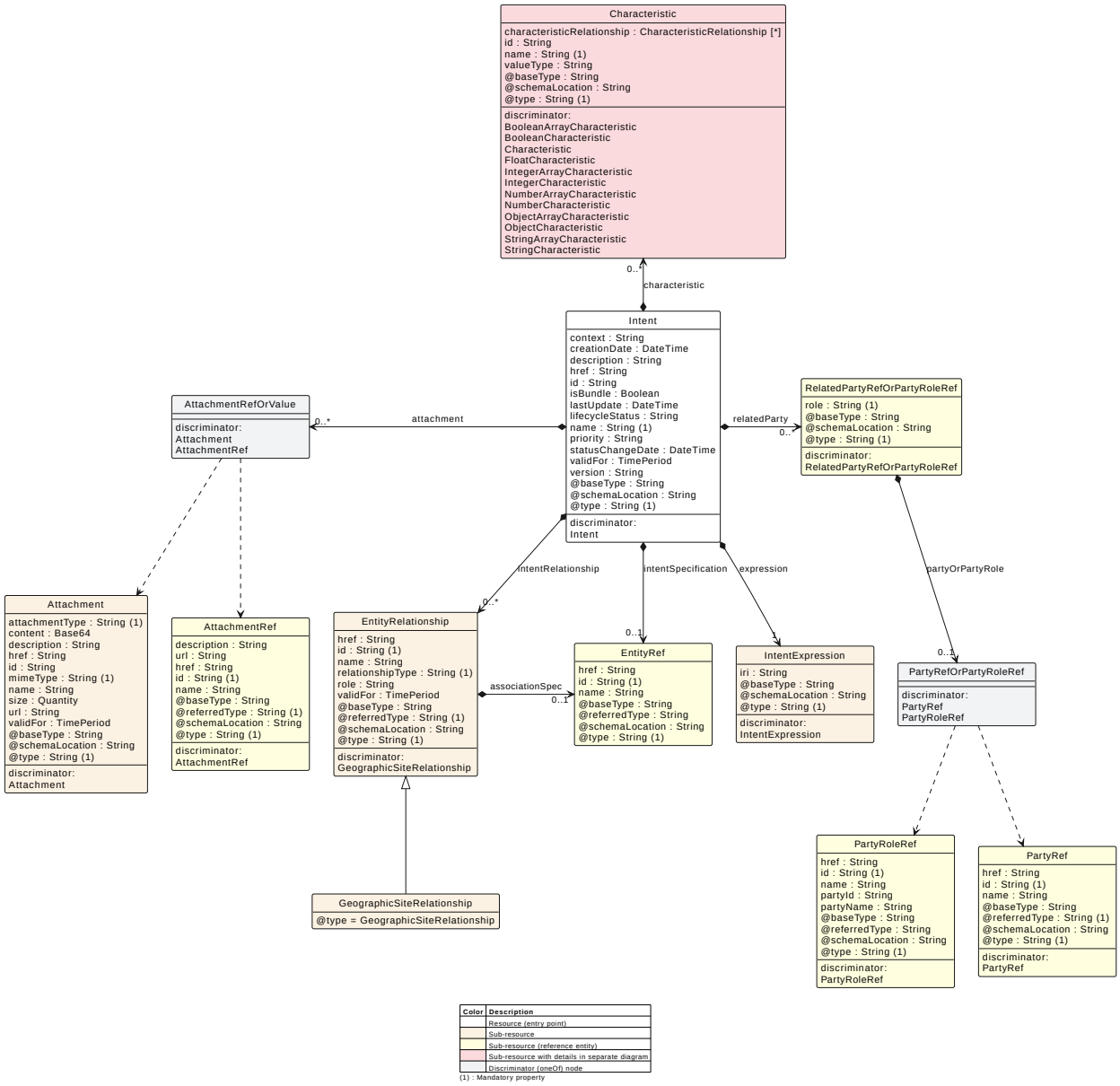



*Figure 20 - Intent*

## Field descriptions

## Customer360 fields

| | |
|---|---|
| account | A Customer360AccountVO [0..*]. List of accounts of a customer. |
| agreement | A Customer360AgreementVO [0..*]. List of agreements of a customer. |
| appointment | A Customer360AppointmentVO [0..*]. List of appointments of a customer. |

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| | |
|---|---|
| customer | A Customer360CustomerVO [1]. Customer value object to contain the important information about the customer. |
| customerBill | A Customer360CustomerBillVO [0..*]. List of bills of a customer. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| loyaltyAccount | A Customer360LoyaltyAccountVO [0..*]. List of loyalty accounts of a customer. |
| partyInteraction | A Customer360PartyInteractionVO [0..*]. List of party interactions of a customer. |
| paymentMethod | A Customer360PaymentMethodVO [0..*]. List of payment methods of a customer. |
| product | A Customer360ProductVO [0..*]. List of products of a customer. |
| productOrder | A Customer360ProductOrderVO [0..*]. List of product orders of a customer. |
| promotion | A Customer360PromotionVO [0..*]. List of promotions of a customer. |
| quote | A Customer360QuoteVO [0..*]. List of quotes of a customer. |
| serviceProblem | A Customer360ServiceProblemVO [0..*]. List of service problems of a customer. |
| troubleTicket | A Customer360TroubleTicketVO [0..*]. List of trouble tickets for a customer. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360AccountVO sub-resource fields**

Acount value object to contain the important information about the accounts of a customer.

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| | |
|---|---|
| accountType | A String. A categorization of an account, such as individual, joint, and so forth, whose instances share some of the same characteristics. Note: for flexibility we use a String here but an implementation may use an enumeration with a limited list of valid values. |
| description | A String. Detailed description of the party account. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. Date of last modification of the account. |
| name | A String [1]. Name of the account. |
| state | A String. Contains the lifecycle state such as: Active, Closed, Suspended and so on. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |
| | Customer360AccountVO can be instantiated as<br>* Customer360LoyaltyAccountVO |
| value | This property is present in subclasses |

**Customer360LoyaltyAccountVO sub-resource fields**

LoyaltyAccount value object to contain the important information about the loyalty account of a customer. A LoyaltyAccount corresponds to a set of balances to carry loyalty movement according to different validity period and unit.

| | |
|---|---|
| accountBalance | An AccountBalance [0..*]. List of balances for the account, for example loyalty earnings wallet, loyalty banked wallet. |
| accountType | A String. A categorization of an account, such as individual, joint, and so forth, whose instances share some of the same characteristics. Note: for flexibility we use a String here but an implementation may use an enumeration with a limited list of valid values. |
| description | A String. Detailed description of the party account. |

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier [0..*]. Collection of external identifiers. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. Date of last modification of the account. |
| name | A String [1]. Name of the account. |
| relatedParty | A RelatedPartyRefOrPartyRoleRef [0..*]. List of parties that have some relationship with the account, for example the loyalty program member. |
| state | A String. Contains the lifecycle state such as: Active, Closed, Suspended and so on. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AccountBalance sub-resource fields**

Balances linked to the account.

| | |
|---|---|
| amount | A Money [1]. A base / value business entity used to represent money. |
| balanceType | A String [1]. Type of the balance : deposit balance, disputed balance, loyalty balance, receivable balance... |
| id | A String. Identifier of the balance within list of the balances (for entities with list). |
| validFor | A TimePeriod [1]. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ExternalIdentifier sub-resource fields**

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An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

| | |
|---|---|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id | A String [1]. Identification of the entity within the external system. |
| owner | A String. Name of the external system that owns the entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**RelatedPartyRefOrPartyRoleRef sub-resource fields**

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

| | |
|---|---|
| partyOrPartyRole | A PartyRefOrPartyRoleRef. |
| role | A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PartyRef sub-resource fields**

A Party reference.

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| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String [1]. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. ENUMERATED with values:<br>* PartyRef |

**PartyRoleRef sub-resource fields**

Party role reference. A party role represents the part played by a party in a given context.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| partyId | A String. The identifier of the engaged party that is linked to the PartyRole object. |
| partyName | A String. The name of the engaged party that is linked to the PartyRole object. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. ENUMERATED with values:<br>* PartyRoleRef |

**Customer360AgreementVO sub-resource fields**

Agreement value object to contain the important information about the agreements of a customer.

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| | |
|---|---|
| agreementPeriod | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| agreementSpecification | An AgreementSpecificationRef. Agreement specification reference. An AgreementSpecification represents a template of an agreement that can be used when establishing partnerships. |
| agreementType | A String. The type of the agreement. For example commercial. |
| engagedParty | A PartyRefOrPartyRoleRef [0..*]. The parties that sign the agreement and are bound by its terms. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String [1]. A human-readable name for the agreement. |
| status | A String. The current status of the agreement. Typical values are: in process, approved and rejected. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AgreementSpecificationRef sub-resource fields**

Agreement specification reference. An AgreementSpecification represents a template of an agreement that can be used when establishing partnerships.

| | |
|---|---|
| description | A String. A narrative that explains in detail what the agreement specification is about. |
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| version | A String. Agreement specification version. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360AppointmentVO sub-resource fields**

Appointment value object to contain the important information about the appointments of a customer.

| | |
|---|---|
| calendarEvent | A CalendarEventRef. Reference of a CalendarEvent. |
| category | A String. Business category : intervention for example or to be more precise after SalesIntervention, orderDelivery Intervention,... |
| creationDate | A DateTime. Appointment creation date. |
| description | A String. Short free text describing the appointment. |
| externalId | A String. External reference known by the Partner. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. Date of last appointment update. |
| status | An AppointmentStateType. Valid values for the lifecycle state of the appointment. ENUMERATED with values:<br>* confirmed<br>* cancelled<br>* pending<br>* rescheduled<br>* inProgress<br>* completed<br>* expired<br>* failed |
| validFor | A TimePeriod [1]. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**CalendarEventRef sub-resource fields**

Reference of a CalendarEvent.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360CustomerVO sub-resource fields**

Customer value object to contain the important information about the customer.

| | |
|---|---|
| contactMedium | A ContactMedium [0..*]. Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContactMedium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |

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| | |
|---|---|
| id | A String. Unique identifier. |
| name | A String. A word, term, or phrase by which the Customer is known and distinguished from other Customers. |
| relatedParty | A RelatedPartyOrPartyRole [0..*]. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity. |
| status | A String. Used to track the lifecycle status of the party role. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ContactMedium sub-resource fields**

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContact Medium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...

| | |
|---|---|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| id | A String. Identifier for this contact medium. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

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| | |
|---|---|
| | ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| value | This property is present in subclasses |

**EmailContactMedium sub-resource fields**

Describes an email that could be used to contact a party (an individual or an organization).

| | |
|---|---|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| emailAddress | A String. Full email address in standard format. |
| id | A String. Identifier for this contact medium. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**FaxContactMedium sub-resource fields**

Describes a fax that could be used to contact a party (an individual or an organization).

| | |
|---|---|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| faxNumber | A String. The fax number of the contact. |
| id | A String. Identifier for this contact medium. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |

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| | |
|---|---|
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicAddressContactMedium sub-resource fields**

Describes a geographical address that could be used to contact a party (an individual or an organization).

| | |
|---|---|
| city | A String. The city. |
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| country | A String. The country. |
| geographicAddress | A GeographicAddressRef. Reference to a Geographic Address. |
| id | A String. Identifier for this contact medium. |
| postCode | A String. Postcode. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |
| stateOrProvince | A String. State or province. |
| street1 | A String. Describes the street. |
| street2 | A String. Complementary street description. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicAddressRef sub-resource fields**

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Reference to a Geographic Address.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PhoneContactMedium sub-resource fields**

Describes a phone number that could be used to contact a party (an individual or an organization).

| | |
|---|---|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| id | A String. Identifier for this contact medium. |
| phoneNumber | A String. The phone number of the contact. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**SocialContactMedium sub-resource fields**

Describes a social media identifier that could be used to contact a party (an individual or an organization).

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| | |
|---|---|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| id | A String. Identifier for this contact medium. |
| preferred | A Boolean. If true, indicates that is the preferred contact medium. |
| socialNetworkId | A String. Identifier as a member of a social network. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**RelatedPartyOrPartyRole sub-resource fields**

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

| | |
|---|---|
| partyOrPartyRole | A PartyOrPartyRole. |
| role | A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**BusinessPartner sub-resource fields**

When business partner is the BusinessPartner.

| | |
|---|---|
| account | An AccountRef [0..*]. Collection of accounts. |
| agreement | An AgreementRef [0..*]. Collection of agreements. |

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| | |
|---|---|
| characteristic | A Characteristic [0..*]. Describes the characteristic of a party role.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| contactMedium | A ContactMedium [0..*]. Collection of contact mediums.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditProfile | A CreditProfile [0..*]. Collection of credit profiles. |
| description | A String. A description of the PartyRole. |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity. |
| partyRoleSpecification | A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context. |
| paymentMethod | A PaymentMethodRef [0..*]. Collection of payment methods. |

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| | |
|---|---|
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| role | A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification. |
| status | A String. Used to track the lifecycle status of the party role. |
| statusReason | A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AccountRef sub-resource fields**

Account reference. A account may be a party account or a financial account.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AgreementRef sub-resource fields**

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Agreement reference. An agreement represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Characteristic sub-resource fields**

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

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| | |
|---|---|
| | Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| value | This property is present in subclasses |

**CharacteristicRelationship sub-resource fields**

Another Characteristic that is related to the current Characteristic;.

| | |
|---|---|
| id | A String [1]. Unique identifier of the characteristic. |
| relationshipType | A String [1]. The type of relationship. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**BooleanArrayCharacteristic sub-resource fields**

A characteristic which value is an array of Boolean(s).

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A Boolean [1..*]. A characteristic which value is an array of Boolean(s). |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**BooleanCharacteristic sub-resource fields**

A characteristic which value is a Boolean.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A Boolean [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**FloatCharacteristic sub-resource fields**

A characteristic which value is a Float.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A Float [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

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### IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | An Integer [1..*]. A characteristic which value is an array of Integer(s). |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

### IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | An Integer [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

### NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

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| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A Number [1..*]. A characteristic which value is an array of Number(s). |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**NumberCharacteristic sub-resource fields**

A characteristic which value is a Number.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A Number [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ObjectArrayCharacteristic sub-resource fields**

A characteristic which value is an array of Object(s).

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |

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| | |
|---|---|
| name | A String [1]. Name of the characteristic. |
| value | A valueType [1..*]. Collection of characteristic values. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ObjectCharacteristic sub-resource fields**

A characteristic which value is a Object.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A valueType_2 [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**StringArrayCharacteristic sub-resource fields**

A characteristic which value is an array of String(s).

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A String [1..*]. Collection of string characteristics. |
| valueType | A String. Data type of the value of the characteristic. |

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| | |
|---|---|
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**StringCharacteristic sub-resource fields**

A characteristic which value is a String.

| | |
|---|---|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id | A String. Unique identifier of the characteristic. |
| name | A String [1]. Name of the characteristic. |
| value | A String [1]. Value of the characteristic. |
| valueType | A String. Data type of the value of the characteristic. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**CreditProfile sub-resource fields**

Credit profile for the party (containing credit scoring, ...). By default only the current credit profile is retrieved. It can be used as a list to give the party credit profiles history, the first one in the list will be the current one.

| | |
|---|---|
| creditProfileDate | A DateTime. The date the profile was established. |
| creditRiskRating | An Integer. This is an integer whose value is used to rate the risk. |
| creditScore | An Integer. A measure of a person or organizations creditworthiness calculated on the basis of a combination of factors such as their income and credit history. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |

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| | |
|---|---|
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PartyRoleSpecificationRef sub-resource fields**

Party role specification reference. A party role specification gives additional details on the part played by a party in a given context.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PaymentMethodRef sub-resource fields**

PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Consumer sub-resource fields**

The part played by a party in a given context.

| | |
|---|---|
| account | An AccountRef [0..*]. Collection of accounts. |
| agreement | An AgreementRef [0..*]. Collection of agreements. |
| characteristic | A Characteristic [0..*]. Describes the characteristic of a party role.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| contactMedium | A ContactMedium [0..*]. Collection of contact mediums.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditProfile | A CreditProfile [0..*]. Collection of credit profiles. |
| description | A String. A description of the PartyRole. |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |

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| | |
|---|---|
| id | A String. Unique identifier. |
| name | A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity. |
| partyRoleSpecification | A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context. |
| paymentMethod | A PaymentMethodRef [0..*]. Collection of payment methods. |
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| role | A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification. |
| status | A String. Used to track the lifecycle status of the party role. |
| statusReason | A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Individual sub-resource fields**

Individual represents a single human being (a man, woman or child). The individual can be a customer, an employee or any other person that the organization needs to store information about.

| | |
|---|---|
| aristocraticTitle | A String. E.g. Baron, Graf, Earl. |
| birthDate | A DateTime. Birth date. |

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| | |
|---|---|
| contactMedium | A ContactMedium [0..*]. List of means for contacting the party, e.g. mobile phone, email address.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| countryOfBirth | A String. Country where the individual was born. |
| creditRating | A PartyCreditProfile [0..*]. List of credit profiles and scores for the party, typically received from an external credit broker. |
| deathDate | A DateTime. Date of death. |
| disability | A Disability [0..*]. List of disabilities suffered by the individual. |
| externalReference | An ExternalIdentifier [0..*]. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system. |
| familyName | A String. Contains the non-chosen or inherited name. Also known as last name in the Western context. |
| familyNamePrefix | A String. Family name prefix. |
| formattedName | A String. A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean). |
| gender | A String. Gender. |
| generation | A String. E.g.. Sr, Jr, III (the third). |
| givenName | A String. First name of the individual. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| individualIdentification | An IndividualIdentification [0..*]. List of official identifications issued to the individual, such as passport, driving licence, social security number. |

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| | |
|---|---|
| languageAbility | A LanguageAbility [0..*]. List of national languages known by the individual. |
| legalName | A String. Legal name or birth name (name one has for official purposes). |
| location | A String. Temporary current location of the individual (may be used if the individual has approved its sharing). |
| maritalStatus | A String. Marital status (married, divorced, widow ...). |
| middleName | A String. Middles name or initial. |
| name | A String. Full name flatten (first, middle, and last names) - this is the name that is expected to be presented in reference data types such as PartyRef, RelatedParty, etc. that refer to Individual. |
| nationality | A String. Nationality. |
| otherName | An OtherNameIndividual [0..*]. List of other names by which this individual is known. |
| partyCharacteristic | A Characteristic [0..*]. List of additional characteristics that a Party can take on.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| placeOfBirth | A String. Reference to the place where the individual was born. |
| preferredGivenName | A String. Contains the chosen name by which the individual prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname. |
| relatedParty | A RelatedPartyOrPartyRole [0..*]. List of parties and/or party roles related to this party. |

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| | |
|---|---|
| skill | A Skill [0..*]. List of skills exhibited by the individual. |
| status | An IndividualStateType. Valid values for the lifecycle state of the individual. ENUMERATED with values:<br>* initialized<br>* validated<br>* deceased |
| taxExemptionCertificate | A TaxExemptionCertificate [0..*]. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax. |
| title | A String. Useful for titles (aristocratic, social,...) Pr, Dr, Sir, ... |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PartyCreditProfile sub-resource fields**

An individual might be evaluated for its worthiness and this evaluation might be based on a credit rating given by a credit agency.

| | |
|---|---|
| creditAgencyName | A String. Name of the credit agency giving the score. |
| creditAgencyType | A String. Type of the credit agency giving the score. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| ratingReference | A String. Reference corresponding to the credit rating. |
| ratingScore | An Integer32. A measure of a party's creditworthiness calculated on the basis of a combination of factors such as their income and credit history. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Disability sub-resource fields**

Lack or inadequate strength or ability.

| | |
|---|---|
| disabilityCode | A String. Code of the disability. |
| disabilityName | A String. Name of the disability. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |

**IndividualIdentification sub-resource fields**

Represents our registration of information used as proof of identity by an individual (passport, national identity card, drivers license, social security number, birth certificate).

| | |
|---|---|
| attachment | An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself. |
| identificationId | A String. Identifier. |
| identificationType | A String. Identification type (passport, national identity card, drivers license, social security number, birth certificate). |
| issuingAuthority | A String. Authority which has issued the identifier, such as: social security, town hall. |
| issuingDate | A DateTime. Date at which the identifier was issued. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Attachment sub-resource fields**

Complements the description of an element (for instance a product) through video,

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pictures...

| | |
|---|---|
| attachmentType | A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy. |
| content | A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url. |
| description | A String. A narrative text describing the content of the attachment. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| mimeType | A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons. |
| name | A String. The name of the attachment. |
| size | A Quantity. An amount in a given unit. |
| url | A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AttachmentRef sub-resource fields**

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

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| | |
|---|---|
| ~~description~~ | DEPRECATED<br>A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6. |
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| ~~url~~ | DEPRECATED<br>A String. Link to the attachment media/content. DEPRECATED: will be removed in v6. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String [1]. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**LanguageAbility sub-resource fields**

Ability of an individual to understand or converse in a language.

| | |
|---|---|
| isFavouriteLanguage | A Boolean. A “true” value specifies whether the language is considered by the individual as his favourite one. |
| languageCode | A String. Language code (RFC 5646). |
| languageName | A String. Language name. |
| listeningProficiency | A String. Listening proficiency evaluated for this language. |
| readingProficiency | A String. Reading proficiency evaluated for this language. |
| speakingProficiency | A String. Speaking proficiency evaluated for this language. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| writingProficiency | A String. Writing proficiency evaluated for this language. |

**OtherNameIndividual sub-resource fields**

Keeps track of other names, for example the old name of a woman before marriage or an artist name.

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| | |
|---|---|
| aristocraticTitle | A String. E.g. Baron, Graf, Earl, etc. |
| familyName | A String. Contains the non-chosen or inherited name. Also known as last name in the Western context. |
| familyNamePrefix | A String. Family name prefix. |
| formattedName | A String. . A fully formatted name in one string with all of its pieces in their proper place and all of the necessary punctuation. Useful for specific contexts (Chinese, Japanese, Korean, etc.). |
| fullName | A String. Full name flatten (first, middle, and last names). |
| generation | A String. E.g. Sr, Jr, etc. |
| givenName | A String. First name. |
| legalName | A String. Legal name or birth name (name one has for official purposes). |
| middleName | A String. Middle name or initial. |
| preferredGivenName | A String. Contains the chosen name by which the person prefers to be addressed. Note: This name may be a name other than a given name, such as a nickname. |
| title | A String. Use for titles (aristrocatic, social, ...): Pr, Dr, Sir,.... |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |

**Skill sub-resource fields**

Skills evaluated for an individual with a level and possibly with a limited validity when an obsolescence is defined (Ex: the first-aid certificate first level is limited to one year and an update training is required each year to keep the level).

| | |
|---|---|
| comment | A String. A free text comment linked to the evaluation done. |
| evaluatedLevel | A String. Level of expertise in a skill evaluated for an individual. |
| skillCode | A String. Code of the skill. |
| skillName | A String. Name of the skill, such as Java language. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |

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### TaxExemptionCertificate sub-resource fields

A tax exemption certificate represents a tax exemption granted to a party (individual or organization) by a tax jurisdiction which may be a city, state, country,... An exemption has a certificate identifier (received from the jurisdiction that levied the tax) and a validity period. An exemption is per tax types and determines for each type of tax what portion of the tax is exempted (partial by percentage or complete) via the tax definition.

| | |
|---|---|
| attachment | An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself. |
| certificateNumber | A String. Identifier of a document that shows proof of exemption from taxes for the taxing jurisdiction. |
| id | A String. Identifier of the tax exemption within list of the exemptions. |
| issuingJurisdiction | A String. Name of the jurisdiction that issued the exemption. |
| reason | A String. Reason for the tax exemption. |
| taxDefinition | A TaxDefinition [0..*]. A list of taxes that are covered by the exemption, e.g. City Tax, State Tax. The definition would include the exemption (e.g. for a rate exemption 0% would be a full exemption, 5% could be a partial exemption if the actual rate was 10%). |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

### TaxDefinition sub-resource fields

Reference of a tax definition. A tax is levied by an authorized tax jurisdiction. For example, there are many different types of tax (Federal Tax levied by the US Government, State Tax levied by the State of California, City Tax levied by the City of Los Angeles, etc.).

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| | |
|---|---|
| id | A String. Unique identifier of the tax. |
| jurisdictionLevel | A String. Level of the jurisdiction that levies the tax. |
| jurisdictionName | A String. Name of the jurisdiction that levies the tax. |
| name | A String. Tax name. |
| taxType | A String. Type of the tax. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Organization sub-resource fields**

Organization represents a group of people identified by shared interests or purpose. Examples include business, department and enterprise. Because of the complex nature of many businesses, both organizations and organization units are represented by the same data.

| | |
|---|---|
| contactMedium | A ContactMedium [0..*]. List of means for contacting the party, e.g. mobile phone, email address.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditRating | A PartyCreditProfile [0..*]. List of credit profiles and scores for the party, typically received from an external credit broker. |
| existsDuring | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| externalReference | An ExternalIdentifier [0..*]. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system. |

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| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isHeadOffice | A Boolean. If value is true, the organization is the head office. |
| isLegalEntity | A Boolean. If value is true, the organization is a legal entity known by a national referential. |
| name | A String. Organization name (department name for example). |
| nameType | A String. Type of the name : Co, Inc, Ltd, etc. |
| organizationChildRelationship | An OrganizationChildRelationship [0..*]. List of organizations that are contained within this organization. For example if this organization is the Legal Department, the child organizations might include Claims, Courts, Contracts. |
| organizationIdentification | An OrganizationIdentification [0..*]. List of official identifiers given to the organization, for example company number in the registry of companies. |
| organizationParentRelationship | An OrganizationParentRelationship. Parent references of an organization in a structure of organizations. |
| organizationType | A String. Type of Organization (company, department...). |
| otherName | An OtherNameOrganization [0..*]. List of additional names by which the organization is known. |
| partyCharacteristic | A Characteristic [0..*]. List of additional characteristics that a Party can take on.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |

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| | |
|---|---|
| relatedParty | A RelatedPartyOrPartyRole [0..*]. List of parties and/or party roles related to this party. |
| status | An OrganizationStateType. Valid values for the lifecycle state of the organization. ENUMERATED with values:<br>* initialized<br>* validated<br>* closed |
| taxExemptionCertificate | A TaxExemptionCertificate [0..*]. List of tax exemptions granted to the party. For example, a war veteran might have partial exemption from state tax and a full exemption from federal tax. |
| tradingName | A String. Name that the organization (unit) trades under. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrganizationChildRelationship sub-resource fields**

Child references of an organization in a structure of organizations.

| | |
|---|---|
| organization | An OrganizationRef. |
| relationshipType | A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrganizationRef sub-resource fields**

Entity reference schema to be use for all entityRef class.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |

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| | |
|---|---|
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrganizationIdentification sub-resource fields**

Represents our registration of information used as proof of identity by an organization.

| | |
|---|---|
| attachment | An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself. |
| identificationId | A String. Identifier. |
| identificationType | A String. Type of identification information used to identify the company in a country or internationally. |
| issuingAuthority | A String. Authority which has issued the identifier (chamber of commerce...). |
| issuingDate | A DateTime. Date at which the identifier was issued. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrganizationParentRelationship sub-resource fields**

Parent references of an organization in a structure of organizations.

| | |
|---|---|
| organization | An OrganizationRef. |

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| | |
|---|---|
| relationshipType | A String. Type of the relationship. Could be juridical, hierarchical, geographical, functional for example. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OtherNameOrganization sub-resource fields**

Keeps track of other names, for example the old name of an organization.

| | |
|---|---|
| name | A String. Organization name (department name for example). |
| nameType | A String. Co. , Inc. , Ltd. , Pty Ltd. , Plc; , Gmbh. |
| tradingName | A String. The name that the organization trades under. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PartyRole sub-resource fields**

The part played by a party in a given context.

| | |
|---|---|
| account | An AccountRef [0..*]. Collection of accounts. |
| agreement | An AgreementRef [0..*]. Collection of agreements. |

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| | |
|---|---|
| characteristic | A Characteristic [0..*]. Describes the characteristic of a party role.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| contactMedium | A ContactMedium [0..*]. Collection of contact mediums.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditProfile | A CreditProfile [0..*]. Collection of credit profiles. |
| description | A String. A description of the PartyRole. |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity. |
| partyRoleSpecification | A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context. |
| paymentMethod | A PaymentMethodRef [0..*]. Collection of payment methods. |

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| | |
|---|---|
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| role | A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification. |
| status | A String. Used to track the lifecycle status of the party role. |
| statusReason | A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |
| | PartyRole can be instantiated as<br>* BusinessPartner<br>* Consumer<br>* Producer<br>* Supplier |
| value | This property is present in subclasses |

**Producer sub-resource fields**

The part played by a party in a given context.

| | |
|---|---|
| account | An AccountRef [0..*]. Collection of accounts. |
| agreement | An AgreementRef [0..*]. Collection of agreements. |

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| | |
|---|---|
| characteristic | A Characteristic [0..*]. Describes the characteristic of a party role.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| contactMedium | A ContactMedium [0..*]. Collection of contact mediums.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditProfile | A CreditProfile [0..*]. Collection of credit profiles. |
| description | A String. A description of the PartyRole. |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity. |
| partyRoleSpecification | A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context. |
| paymentMethod | A PaymentMethodRef [0..*]. Collection of payment methods. |

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| | |
|---|---|
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| role | A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification. |
| status | A String. Used to track the lifecycle status of the party role. |
| statusReason | A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Supplier sub-resource fields**

When business partner is the Supplier.

| | |
|---|---|
| account | An AccountRef [0..*]. Collection of accounts. |
| agreement | An AgreementRef [0..*]. Collection of agreements. |

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| | |
|---|---|
| characteristic | A Characteristic [0..*]. Describes the characteristic of a party role.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| contactMedium | A ContactMedium [0..*]. Collection of contact mediums.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creditProfile | A CreditProfile [0..*]. Collection of credit profiles. |
| description | A String. A description of the PartyRole. |
| engagedParty | A PartyRef [1]. A Party reference. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String [1]. A word, term, or phrase by which the PartyRole is known and distinguished from other PartyRoles. It's the name of the PartyRole unique entity. |
| partyRoleSpecification | A PartyRoleSpecificationRef. Party role specification reference. A party role specification gives additional details on the part played by a party in a given context. |
| paymentMethod | A PaymentMethodRef [0..*]. Collection of payment methods. |

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| | |
|---|---|
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| role | A String. Role played by the engagedParty in this context. As role is defined by partyRoleSpecification, this role attribute can be used to precise the role defined by party RoleSpecification, or it can be used to define the role in case there is no partyRoleSpecification. |
| status | A String. Used to track the lifecycle status of the party role. |
| statusReason | A String. A string providing an explanation on the value of the status lifecycle. For instance if the status is Rejected, statusReason will provide the reason for rejection. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360CustomerBillVO sub-resource fields**

CustomerBill value object to contain the important information about the bills of a customer.

| | |
|---|---|
| billNo | A String. Bill reference known by the customer or the party and displayed on the bill. Could be different from the id. |
| category | A String. Category of the bill produced : normal, duplicate, interim, last, trial customer or credit note for example. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |

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| | |
|---|---|
| state | A CustomerBillStateType. Recommended Enumeration Type (not formal forced in standard): Valid values for the lifecycle state of the bill: new = 'bill is ready to validate or to sent', validated = 'bill is checked (manual / automatic)', sent = 'bill is sent with the channel defined in the billingaccount', settled = 'bill is payed', partiallySettled = 'bill is partially payed', onHold = 'bill will not be in further processing until open issues connected to the bill are solved'. ENUMERATED with values:<br>* new<br>* onHold<br>* validated<br>* sent<br>* settled<br>* partiallyPaid |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360PartyInteractionVO sub-resource fields**

PartyInteraction value object to contain the important information about the party interactions of a customer.

| | |
|---|---|
| description | A String. Description of the interaction. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| interactionDate | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| reason | A String. Reason why the interaction happened. |
| status | A String. Status of the interaction (opened, inProgress, completed). |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360PaymentMethodVO sub-resource fields**

PaymentMethod value object to contain the important information about the payment methods of a customer.

| | |
|---|---|
| description | A String. The contents of the payment method. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| isPreferred | A Boolean. If the method is the preferred one by the owner. Typically used when querying for the payment methods of a specific customer or account. |
| name | A String [1]. Friendly name assigned to the payment method. |
| status | A String [1]. |
| statusDate | A DateTime. The date when the status was set. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360ProductVO sub-resource fields**

Product value object to contain the important information about the products of a customer.

| | |
|---|---|
| description | A String. Is the description of the product. It could be copied from the description of the Product Offering. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. Name of the product. It could be the same as the name of the product offering. |

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| | |
|---|---|
| productOffering | A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information. |
| status | A ProductStatusType. Possible values for the status of the product. ENUMERATED with values:<br>* created<br>* pendingActive<br>* cancelled<br>* active<br>* pendingTerminate<br>* terminated<br>* suspended<br>* aborted |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductOfferingRef sub-resource fields**

ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| version | A String. Version of the product offering. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360ProductOrderVO sub-resource fields**

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ProductOrder value object to contain the important information about the product orders of a customer.

| | |
|---|---|
| category | A String. Used to categorize the order from a business perspective that can be useful for the OM system (e.g. "enterprise", "residential", ...). |
| completionDate | A DateTime. Date when the order was completed. |
| creationDate | A DateTime. Date when the order was created. |
| description | A String. Description of the product order. |
| expectedCompletionDate | A DateTime. Expected delivery date amended by the provider. |
| externalId | A String. ID given by the consumer and only understandable by him (to facilitate his searches afterwards). |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| priority | A String. A way that can be used by consumers to prioritize orders in OM system (from 0 to 4 : 0 is the highest priority, and 4 the lowest). |
| productOrderItem | A ProductOrderItem [0..*]. Collection of product order items. |
| state | A ProductOrderStateType [1]. Possible values for the state of the order. ENUMERATED with values:<br>* acknowledged<br>* rejected<br>* pending<br>* held<br>* inProgress<br>* cancelled<br>* completed<br>* failed<br>* partial<br>* assessingCancellation<br>* pendingCancellation<br>* draft<br>* inProgress.accepted |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductOrderItem sub-resource fields**

An identified part of the order. A product order is decomposed into one or more order items.

| | |
|---|---|
| action | An ItemActionType [1]. Action to be performed on the entity managed by the item. ENUMERATED with values:<br>* add<br>* modify<br>* delete<br>* noChange |
| appointment | An AppointmentRef. Refers an appointment, such as a Customer presentation or internal meeting or site visit. |
| billingAccount | A BillingAccountRef. BillingAccount reference. A BillingAccount is a detailed description of a bill structure. |
| id | A String [1]. Identifier of the ProductOrder item (generally it is a sequence number 01, 02, 03, ...). |
| itemPrice | An OrderPrice [0..*]. Collection of product order item prices. |
| itemTerm | An OrderTerm [0..*]. Collection of product order item terms. |
| itemTotalPrice | An OrderPrice [0..*]. Collection of order item prices. |
| note | A Note [0..*]. Collection of order item notes. |
| payment | A PaymentRef [0..*]. Collection of order item payments. |
| product | A ProductRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Product entity and not the ProductRefOrValue class itself. |
| productOffering | A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information. |

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| | |
|---|---|
| productOfferingQualificationItem | A ProductOfferingQualificationItemRef. It's a productOfferingQualification item that has been executed previously. |
| productOrderItem | A ProductOrderItem [0..*]. Collection of order items. |
| productOrderItemRelationship | An OrderItemRelationship [0..*]. Collection of order items relationships. |
| qualification | A ProductOfferingQualificationRef [0..*]. Collection of order item qualifications. |
| quantity | An Integer. Quantity ordered. |
| quoteItem | A QuoteItemRef. A reference to Quote item that has been executed previously. |
| state | A ProductOrderItemStateType [1]. Possible values for the state of the product order item. ENUMERATED with values:<br>* acknowledged<br>* rejected<br>* pending<br>* held<br>* inProgress<br>* cancelled<br>* completed<br>* failed<br>* partial<br>* assessingCancellation<br>* pendingCancellation |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AppointmentRef sub-resource fields**

Refers an appointment, such as a Customer presentation or internal meeting or site visit.

| | |
|---|---|
| description | A String. An explanatory text regarding the appointment made with a party. |
| href | A String. The URI of the referred entity. |

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| | |
|---|---|
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**BillingAccountRef sub-resource fields**

BillingAccount reference. A BillingAccount is a detailed description of a bill structure.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| ratingType | A String. Indicates whether the account follows a specific payment option such as prepaid or postpaid. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrderPrice sub-resource fields**

An amount, usually of money, that represents the actual price paid by the Customer for this item or this order.

| | |
|---|---|
| billingAccount | A BillingAccountRef. BillingAccount reference. A BillingAccount is a detailed description of a bill structure. |
| description | A String. A narrative that explains in detail the semantics of this order item price. |

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| | |
|---|---|
| name | A String. A short descriptive name such as "Subscription price". |
| price | A Price [1]. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration. |
| priceAlteration | A PriceAlteration [0..*]. A strucuture used to describe a price alteration. |
| priceType | A String [1]. Indicate if the price is for recurrent or no-recurrent charge. |
| productOfferingPrice | A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased. |
| recurringChargePeriod | A String. Could be month, week... |
| unitOfMeasure | A String. Could be minutes, GB... |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Price sub-resource fields**

Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration.

| | |
|---|---|
| dutyFreeAmount | A Money. A base / value business entity used to represent money. |
| percentage | A Float. Percentage to apply for ProdOfferPriceAlteration. |
| taxIncludedAmount | A Money. A base / value business entity used to represent money. |
| taxRate | A Float. Tax rate. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PriceAlteration sub-resource fields**

Is an amount, usually of money, that modifies the price charged for an order item.

| | |
|---|---|
| applicationDuration | An Integer. Duration during which the alteration applies on the order item price (for instance 2 months free of charge for the recurring charge). |
| description | A String. A narrative that explains in detail the semantics of this order item price alteration. |
| name | A String. Name of the order item price alteration. |
| price | A Price [1]. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration. |
| priceType | A String [1]. A category that describes the price such as recurring, one time and usage. |
| priority | An Integer. Priority level for applying this alteration among all the defined alterations on the order item price. |
| productOfferingPrice | A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased. |
| recurringChargePeriod | A String. Could be month, week... |
| unitOfMeasure | A String. Could be minutes, GB... |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductOfferingPriceRef sub-resource fields**

ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased.

| | |
|---|---|
| href | A String. The URI of the referred entity. |

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| | |
|---|---|
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| version | A String. Version of the product offering price. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrderTerm sub-resource fields**

Description of a productTerm linked to this orderItem. This represent a commitment with a duration.

| | |
|---|---|
| description | A String. Description of the productOrderTerm. |
| duration | A Duration. A time interval in a given unit of time. |
| name | A String. Name of the productOrderTerm. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Note sub-resource fields**

Extra information about a given entity.

| | |
|---|---|
| author | A String. Author of the note. |
| date | A DateTime. Date of the note. |
| id | A String. Identifier of the note within its containing entity. |
| text | A String. Text of the note. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PaymentRef sub-resource fields**

If an immediate payment has been done at the product order submission, the payment information are captured and stored (as a reference) in the order.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Product sub-resource fields**

A product offering procured by a customer or other interested party playing a party role. A product is realized as one or more service(s) and / or resource(s).

| | |
|---|---|
| agreementItem | An AgreementItemRef [0..*]. Collection of agreements. |
| billingAccount | A BillingAccountRef. BillingAccount reference. A BillingAccount is a detailed description of a bill structure. |
| creationDate | A DateTime. Date and time when the product was created. |
| description | A String. Is the description of the product. It could be copied from the description of the Product Offering. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| intent | An IntentRefOrValue. Intent Ref (if Intent already exists) or Value (if Intent be created or its details be presented). |

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| | |
|---|---|
| isBundle | A Boolean. If true, the product is a ProductBundle which is an instantiation of a BundledProductOffering. If false, the product is a ProductComponent which is an instantiation of a SimpleProductOffering. |
| isCustomerVisible | A Boolean. If true, the product is visible by the customer. |
| name | A String. Name of the product. It could be the same as the name of the product offering. |
| orderDate | A DateTime. Is the date when the product was ordered. |
| place | A RelatedPlaceRefOrValue [0..*]. Collection of places. |
| product | A ProductRefOrValue [0..*]. Collection of products. |
| productCharacteristic | A Characteristic [0..*]. Collection of product characteristics.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| productOffering | A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information. |
| productOrderItem | A RelatedOrderItem [0..*]. Collection of product order items. |
| productPrice | A ProductPrice [0..*]. Collection of product prices. |
| productRelationship | A ProductRelationship [0..*]. Collection of product relationships. |
| productSerialNumber | A String. Is the serial number for the product. This is typically applicable to tangible products e.g. Broadband Router. |

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| | |
|---|---|
| productSpecification | A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog. |
| productTerm | A ProductTerm [0..*]. Collection of product terms. |
| realizingResource | A ResourceRef [0..*]. Collection of realizing resources. |
| realizingService | A ServiceRef [0..*]. Collection of realizing services. |
| relatedParty | A RelatedPartyOrPartyRole [0..*]. Collection of related parties. |
| startDate | A DateTime. Is the date from which the product starts. |
| status | A ProductStatusType. Possible values for the status of the product. ENUMERATED with values:<br>* created<br>* pendingActive<br>* cancelled<br>* active<br>* pendingTerminate<br>* terminated<br>* suspended<br>* aborted |
| terminationDate | A DateTime. Is the date when the product was terminated. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**AgreementItemRef sub-resource fields**

It's a Agreement item that has been executed previously.

| | |
|---|---|
| agreementHref | A String. Reference of the related entity. |
| agreementId | A String [1]. Unique identifier of a related entity. |
| agreementItemId | A String [1]. Id of an item of a Agreement. |
| agreementName | A String. Name of the related entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Intent sub-resource fields**

An Intent instance is the formal description of all expectations including requirements, goals, and constraints given to a technical system.

| | |
|---|---|
| attachment | An AttachmentRefOrValue [0..*]. Attachments that may be of relevance to this intent, such as picture, document, media. |
| characteristic | A Characteristic [0..*]. Collection of characteristics.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| context | A String. A string used to give a context to the intent. |
| creationDate | A DateTime. Date and time of the creation of this REST resource. |
| description | A String. The description of the intent. |
| expression | An IntentExpression [1]. An Intent expression. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| intentRelationship | An EntityRelationship [0..*]. A list of intents related to this intent. |

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| | |
|---|---|
| intentSpecification | An EntityRef. Entity reference schema to be use for all entityRef class. |
| isBundle | A Boolean. IsBundle determines whether an intent represents a single intent (false), or a bundle of intents(true). |
| lastUpdate | A DateTime. Date and time of the last update of this REST resource. |
| lifecycleStatus | A String. Used to indicate the current lifecycle status of this intent. |
| name | A String [1]. The name of the intent. |
| priority | A String. Can be used by intent owner to prioritize intents in an intent management system. |
| relatedParty | A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties. |
| statusChangeDate | A DateTime. A date time( DateTime). The date that the entity status changed to the current one. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| version | A String. A field that identifies the specific version of an instance of an intent. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**IntentExpression sub-resource fields**

An Intent expression.

| | |
|---|---|
| iri | A String. Internationalized Resource Identifier of the intent Expression. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**EntityRelationship sub-resource fields**

A uni-directional relationship from this entity to a target entity instance.

| | |
|---|---|
| associationSpec | An EntityRef. Entity reference schema to be use for all entityRef class. |
| href | A String. |
| id | A String [1]. |
| name | A String. |
| relationshipType | A String [1]. Type of relationship such as migration, substitution, dependency, exclusivity. |
| role | A String. The association role for this entity. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String [1]. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. |
| | EntityRelationship can be instantiated as<br>* GeographicSiteRelationship |
| value | This property is present in subclasses |

**EntityRef sub-resource fields**

Entity reference schema to be use for all entityRef class.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicSiteRelationship sub-resource fields**

Details of a geographic site relationship.

| | |
|---|---|
| associationSpec | An EntityRef. Entity reference schema to be use for all entityRef class. |
| href | A String. |
| id | A String [1]. |
| name | A String. |
| relationshipType | A String [1]. Type of relationship between GeographicSites, such as adjacentTo, connectedBy. |
| role | A String. The association role for this entity. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String [1]. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. |

**IntentRef sub-resource fields**

Intent reference, for when Intent is used by other entities.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**RelatedPlaceRefOrValue sub-resource fields**

Use `role` to distinguish the context of the Place being referenced (such as a `delivery-entrance`). In the context of a Ref, use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). In the context of a Value, use `@type` to specify the Place sub-type.

| | |
|---|---|
| place | A PlaceRefOrValue [1]. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself. |
| role | A String [1]. The context in which this Place is being used. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicAddress sub-resource fields**

Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanProperty Address.

| | |
|---|---|
| city | A String. City that the address is in. |
| country | A String. Country that the address is in. |
| countryCode | A StandardIdentifier [0..*]. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes. |

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| | |
|---|---|
| description | A String. Additional information of a user-friendly description of the Place. |
| externalIdentifier | An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers. |
| geographicAddressRelationship | A GeographicAddressRelationship [0..*]. A list of geographic addresses relationships. Represents a relationship between geographic addresses. |
| geographicAddressType | A String. Classification of the address, e.g., residential, industrial. |
| geographicLocation | A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself. |
| geographicSite | A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself. |
| geographicSubAddress | A GeographicSubAddress [0..*]. Collection of sub addresses. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| locality | A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET]. |
| name | A String. A user-friendly name for the Place. |
| postcode | A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode). |
| stateOrProvince | A String. The State or Province that the address is in. |
| streetName | A String. Name of the street or other street type. |
| streetNr | A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses. |

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| | |
|---|---|
| streetNrLast | A String. Last number in a range of street numbers allocated to a property. |
| streetNrLastSuffix | A String. Last street number suffix for a ranged address. |
| streetNrSuffix | A String. The first street number suffix. |
| streetSuffix | A String. A modifier denoting a relative direction. |
| streetType | A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**StandardIdentifier sub-resource fields**

The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource. It is anticipated that multiple standards can provide definitions for a single entity, e.g., a country identifier can be specified in various standards (e.g., "ISO 3166-1 Alpha 2", "ISO 3166-1 Alpha 3", "ISO 3166-1 Numeric").

| | |
|---|---|
| format | A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| value | A String. The value of the resource in the corresponding standard.e.g., a country code value. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicAddressRelationship sub-resource fields**

The GeographicAddressRelationship schema represents a relationship between geographic

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addresses. It defines the structure for storing information about how two geographic addresses are related to each other within a system.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. Unique identifier of the related geographic address. This identifier serves as a reference to uniquely identify the associated geographic address within the system. |
| name | A String. Name of the geographic address. This attribute provides a human-readable name or label for the related geographic address, which can help users identify and differentiate between different addresses. |
| relationshipType | A String [1]. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicLocation sub-resource fields**

A GeographicLocation is a pure-virtual super-class to the GeoJSON-aligned geometries of Point (addresses and locations), MultiPoint, LineString (streets, highways and boundaries), MultiLineString and Polygon (countries, provinces, tracts of land). Use the @type attribute to specify which of these is being specified by the geometry attribute.

| | |
|---|---|
| bbox | A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry. |
| description | A String. Additional information of a user-friendly description of the Place. |

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| | |
|---|---|
| externalIdentifier | An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| name | A String. A user-friendly name for the Place. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicLocationRef sub-resource fields**

Reference to a Geographic Location.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicSite sub-resource fields**

A GeographicSite is a managed place, such as a retail store, warehouse or data centre. It will have related responsible parties, a calendar of access times and be described by addresses and/or locations.

| | |
|---|---|
| calendar | A CalendarPeriod [0..*]. Periods of (for example) access, opening, alarm, unavailability of this site. |

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| | |
|---|---|
| capacity | A Quantity [0..*]. Capacities for the site indicating maximum operational thresholds (for example: people, floorspace, racks). |
| ~~code~~ | DEPRECATED<br>A String. Deprecated: Use id, externalIdentifier and/or site Category or name. |
| contactMedium | A ContactMedium [0..*]. Contact information for people with responsibility for this site.<br><br>ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| creationDate | A DateTime. Date and time when the GeographicSite was created. |
| description | A String. Additional information of a user-friendly description of the Place. |
| externalIdentifier | An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers. |
| geographicSiteRelationship | A GeographicSiteRelationship [0..*]. This site might have relationships to other sites. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. Date and time when the GeographicSite was last updated. |
| name | A String [1]. A user-friendly name for the Place. |
| note | A Note [0..*]. Unstructured notes made about this site. |
| relatedAddress | A RelatedGeographicAddressRef [0..*]. Collection of related address references (such as delivery-address, office-address). |
| relatedLocation | A RelatedGeographicLocationRef [0..*]. Collection of related location references (such as retail-complex, service-access). |

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| | |
|---|---|
| relatedParty | A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related roles and parties references (such as site manager, site owner, site operator). |
| siteCategory | A String. Site classification/category. |
| siteFeature | A GeographicSiteFeature [0..*]. Collection of site features. |
| status | A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**CalendarPeriod sub-resource fields**

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

| | |
|---|---|
| day | A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values:<br>* mon<br>* tue<br>* wed<br>* thu<br>* fri<br>* sat<br>* sun<br>* weekdays<br>* weekend |
| hourPeriod | A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format. |
| status | A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.). |

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| | |
|---|---|
| timeZone | A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**HourPeriod sub-resource fields**

Hour interval.

| | |
|---|---|
| endHour | A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight). |
| startHour | A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM). |

**RelatedGeographicAddressRef sub-resource fields**

A Postal Address (by reference) with its associated role.

| | |
|---|---|
| address | A GeographicAddressRef [1]. Reference to a Geographic Address. |
| role | A String [1]. The context in which this Address is being used. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**RelatedGeographicLocationRef sub-resource fields**

A Geographic Location (by reference) with its associated role.

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| | |
|---|---|
| location | A GeographicLocationRef [1]. Reference to a Geographic Location. |
| role | A String [1]. The context in which this location is being used. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicSiteFeature sub-resource fields**

Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.

| | |
|---|---|
| attachment | An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation. |
| featureCategory | A String [0..*]. Collection of feature categories. |
| featureCharacteristic | A Characteristic [0..*]. This is a list of Characteristics for a particular feature.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| featureRelationship | A FeatureRelationship [0..*]. Collection of feature relationships. |
| id | A String. Unique identifier. |
| isBundle | A Boolean. True if this is a feature group. Default is false. |
| isEnabled | A Boolean. True if this feature is enabled. Default is true. |

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| | |
|---|---|
| name | A String [1]. The name for the feature. |
| note | A Note [0..*]. Notes relevant to this Feature. |
| policyConstraint | A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints. |
| relatedParty | A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature. |
| validFor | A CalendarPeriod [0..*]. Validity period of this Feature. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**FeatureRelationship sub-resource fields**

Configuration feature.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. This is the name of the target feature. |
| relationshipType | A String [1]. This is the type of the feature relationship. ENUMERATED with values:<br>* excluded<br>* includes<br>* may include<br>* requires |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

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**PolicyRef sub-resource fields**

Reference to managed Policy object.

| | |
|---|---|
| href | A String. Hyperlink reference. |
| id | A String [1]. Unique identifier. |
| name | A String. Name of the referred entity. |
| version | A String. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Feature sub-resource fields**

Configuration feature.

| | |
|---|---|
| attachment | An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation. |
| featureCategory | A String [0..*]. Collection of feature categories. |
| featureCharacteristic | A Characteristic [0..*]. This is a list of Characteristics for a particular feature.<br><br>Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |

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| | |
|---|---|
| featureRelationship | A FeatureRelationship [0..*]. Collection of feature relationships. |
| id | A String. Unique identifier. |
| isBundle | A Boolean. True if this is a feature group. Default is false. |
| isEnabled | A Boolean. True if this feature is enabled. Default is true. |
| name | A String [1]. The name for the feature. |
| note | A Note [0..*]. Notes relevant to this Feature. |
| policyConstraint | A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints. |
| relatedParty | A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature. |
| validFor | A CalendarPeriod [0..*]. Validity period of this Feature. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |
| | Feature can be instantiated as<br>* GeographicSiteFeature |
| value | This property is present in subclasses |

**GeographicSiteRef sub-resource fields**

Entity reference schema to be use for all entityRef class.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

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| | |
|---|---|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicSubAddress sub-resource fields**

Representation of a GeographicSubAddress It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.

| | |
|---|---|
| buildingName | A String. Allows for buildings that have well-known names. |
| href | A String. Link to the subAddress. |
| id | A String. Unique Identifier of the subAddress. |
| levelNumber | A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2. |
| levelType | A String. Describes level types within a building. |
| name | A String. Name of the subAddress to identify it with a meaningful identification. |
| privateStreetName | A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office. |
| privateStreetNumber | A String. Private streets numbers internal to a private street. |
| subAddressType | A String. Type of subAddress : it can be a subunit or a private street. |
| subUnit | A GeographicSubAddressUnit [0..*]. Representation of a SubUnit. It is used for describing subunit within a sub Address e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**GeographicSubAddressUnit sub-resource fields**

Representation of a SubUnit. It is used for describing subunit within a subAddress e.g.

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BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.

| | |
|---|---|
| subUnitNumber | A String [1]. The discriminator used for the subunit, often just a simple number but may also be a range. |
| subUnitType | A String [1]. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**PlaceRef sub-resource fields**

A reference to a physical Place.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The specific sub-type of Place being referred to (such as a `GeographicAddress`). |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductRef sub-resource fields**

A Product reference.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**RelatedOrderItem sub-resource fields**

It's a Order item that has been executed previously.

| | |
|---|---|
| orderHref | A String. Reference of the related entity. |
| orderId | A String [1]. Unique identifier of a related Order. |
| orderItemAction | An ItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values:<br>* add<br>* modify<br>* delete<br>* noChange |
| orderItemId | A String [1]. Id of an item of a product order. |
| role | A String [1]. Role played by the Order. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductPrice sub-resource fields**

Description of price and discount awarded.

| | |
|---|---|
| description | A String. Description of the Product price. |
| name | A String. Name of the Product price. |
| price | A Price [1]. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration. |
| priceAlteration | A PriceAlteration [0..*]. Collection of price changes. |

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| | |
|---|---|
| priceType | A String [1]. Indicate if the price is for recurrent or no-recurrent charge. |
| productOfferingPrice | A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased. |
| recurringChargePeriod | A String. Used for recurring charge to indicate period (month, week, etc..). |
| unitOfMeasure | A String. Unit of Measure if price depending on it (Gb, SMS volume, etc..). |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductRelationship sub-resource fields**

Used to describe relationship between product.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. Id of the related product. |
| name | A String. Name of the referred entity. |
| relationshipType | A String [1]. Relationship type as relies on, bundles, etc... |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductSpecificationRef sub-resource fields**

ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.

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| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| targetProductSchema | A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification. |
| version | A String. Version of the product specification. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**TargetProductSchema sub-resource fields**

The reference object to the schema and type of target product which is described by product specification.

| | |
|---|---|
| @schemaLocation | A Uri [1]. This field provides a link to the schema describing the target product. |
| @type | A String [1]. Class type of the target product. |

**ProductTerm sub-resource fields**

Description of a productTerm linked to this product. This represent a commitment with a duration.

| | |
|---|---|
| description | A String. Description of the productTerm. |
| duration | A Duration. A time interval in a given unit of time. |
| name | A String. Name of the productTerm. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |

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| | |
|---|---|
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ResourceRef sub-resource fields**

Resource reference, for when Resource is used by other entities.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ServiceRef sub-resource fields**

Service reference, for when Service is used by other entities.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductOfferingQualificationItemRef sub-resource fields**

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It's a productOfferingQualification item that has been executed previously.

| | |
|---|---|
| itemId | A String [1]. Id of an item of a product offering qualification. |
| productOfferingQualificationHref | A String. Reference of the related entity. |
| productOfferingQualificationId | A String [1]. Unique identifier of a related entity. |
| productOfferingQualificationName | A String. Name of the related entity. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**OrderItemRelationship sub-resource fields**

Used to describe relationship between Order item. These relationship could have an impact on pricing and conditions.

| | |
|---|---|
| id | A String [1]. Id of the related Order item (must be in the same Order). |
| relationshipType | A String [1]. Relationship type as relies on, bundles, etc... |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**ProductOfferingQualificationRef sub-resource fields**

A productOfferingQualification that has been executed previously.

| | |
|---|---|
| href | A String. The URI of the referred entity. |
| id | A String [1]. The identifier of the referred entity. |
| name | A String. Name of the referred entity. |

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| | |
|---|---|
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**QuoteItemRef sub-resource fields**

A reference to Quote item that has been executed previously.

| | |
|---|---|
| quoteHref | A String. Reference of the related quote. |
| quoteId | A String [1]. Unique identifier of a refered quote. |
| quoteItemId | A String [1]. Identifier of a quote item. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @referredType | A String. The actual type of the target instance when needed for disambiguation. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360PromotionVO sub-resource fields**

Promotion value object to contain the important information about the promotions of a customer.

| | |
|---|---|
| description | A String. Description of Promotion. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. Latest update date of Promotion. |
| lifecycleStatus | A String. Status of Promotion, including draft/Test/WaitForApproval/Release/Suspend/Retirement. |
| name | A String [1]. Name of Promotion. |

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| | |
|---|---|
| type | A String. Type of promotion.The basic type is Award/ Discount/Reduction. More types can be extended in future. |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360QuoteVO sub-resource fields**

Quote value object to contain the important information about the quotes of a customer.

| | |
|---|---|
| category | A String. Used to categorize the quote from a business perspective that can be useful for the CRM system (e.g. "enterprise", "residential", ...). |
| creationDate | A DateTime. Date and time when the quote was created. |
| description | A String. Description of the quote. |
| effectiveQuoteCompletionDate | A DateTime. Date when the quote has been completed. |
| expectedFulfillmentStartDate | A DateTime. This is the date wished by the requester to have the requested quote item delivered. |
| expectedQuoteCompletionDate | A DateTime. This is expected date - from quote supplier - to be able to send back a response for this quote. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |

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| | |
|---|---|
| state | A QuoteStateType. Possible values for the state of the quote. ENUMERATED with values:<br>* acknowledged<br>* rejected<br>* pending<br>* inProgress<br>* cancelled<br>* approved<br>* accepted<br>* declined |
| validFor | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| version | A String. Quote version - if the Partner rejected the quote but negotiations still open a new version of the quote is managed. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360ServiceProblemVO sub-resource fields**

ServiceProblem value object to contain the important information about the service problems of a customer.

| | |
|---|---|
| affectedService | A ServiceRef [0..*]. List of affected services. At least one of affectedResource, affectedService or affectedLocation should be present. |
| category | A String. Classifier for the problem. Settable. For example, this is used for distinguish the category of problem originator in [role].[category] format. Example: service Provider.declarer, supplier.originated, system.originated. |
| description | A String. Free form text describing the Service Problem. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| originatingSystem | A String. Indicates where the problem was generated. |

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| | |
|---|---|
| priority | An Integer. An indication varying from 1 (highest) to 10 (lowest) of how important it is for the service provider to correct the Service Problem. |
| reason | A String. Reason, if known, for the service problem that will assist the Service Provider in understanding the underlying cause or contributing factors, enabling more accurate diagnosis and resolution. This can be set to Unknown, otherwise. |
| status | A String. The current status of the service problem. Possible values are Submitted, Rejected, Acknowledged, In Progress [Held, Pending], Resolved, Closed, and Cancelled. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Customer360TroubleTicketVO sub-resource fields**

TroubleTicket value object to contain the important information about the trouble tickets for a customer.

| | |
|---|---|
| creationDate | A DateTime. The date on which the trouble ticket was created. |
| description | A String. Description of the trouble or issue. |
| expectedResolutionDate | A DateTime. The expected resolution date determined by the trouble ticket system. |
| externalIdentifier | A String. Additional identifier coming from an external system. |
| href | A String. Hyperlink reference. |
| id | A String. Unique identifier. |
| lastUpdate | A DateTime. The date and time that the trouble ticked was last updated. |
| name | A String. Name of the trouble ticket, typically a short description provided by the user that create the ticket. |

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| | |
|---|---|
| priority | A String. The priority of the trouble ticket and how quickly the issue should be resolved. Example: Critical, High, Medium, Low. The value is set by the ticket management system considering the severity, ticket type etc... |
| requestedResolutionDate | A DateTime. The resolution date requested by the user. |
| severity | A String. The severity of the issue. Indicate the implication of the issue on the expected functionality e.g. of a system, application, service etc.. Severity values can be for example : Critical, Major, Minor. |
| status | A TroubleTicketStatusType. Possible values for the status of the trouble ticket. ENUMERATED with values:<br>* acknowledged<br>* rejected<br>* pending<br>* held<br>* inProgress<br>* cancelled<br>* closed<br>* resolved |
| statusChangeDate | A DateTime. The date and time the status changed. |
| statusChangeReason | A String. The reason for changing the status. |
| ticketType | A String. Represent a business type of the trouble ticket e.g. incident, complain, request. |
| @baseType | A String. When sub-classing, this defines the super-class. |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |

**Json representation sample(s)**

Example of a Customer360 resource with complete customer information


```
{
  "@type": "Customer360",
  "@schemaLocation": "/schema-registry/schema/Customer360.yaml",
  "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
  "id": "1140",
  "validFor": {
    "startDateTime": "2018-06-13T00:00:00.000Z",
```


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```
      "endDateTime": "2019-01-11T00:00:00.000Z"
    },
    "customer": {
      "@type": "Customer360CustomerVO",
      "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
      "id": "1140",
      "contactMedium": [
        {
          "@type": "PhoneContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/schema/PhoneContactMedium",
          "id": "2256789876",
          "preferred": true,
          "contactType": "homePhone",
          "validFor": {
            "startDateTime": "2018-06-13T00:00:00.000Z",
            "endDateTime": "2019-01-11T00:00:00.000Z"
          },
          "phoneNumber": "01 09 75 83 51"
        },
        {
          "@type": "GeographicAddressContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/schema/GeographicAddressContactMedium",
          "id": "12345",
          "preferred": false,
          "stateOrProvince": "NY",
          "street2": "123 Lindsay Street",
          "contactType": "homeAddress",
          "validFor": {
            "startDateTime": "2018-06-13T00:00:00.000Z",
            "endDateTime": "2019-01-11T00:00:00.000Z"
          },
          "geographicAddress": {
            "@type": "GeographicAddressRef",
            "href": "https://host:port/tmf-
api/geographicAddressManagement/v5/geographicAddress/12345",
            "id": "12345",
            "@referredType": "GeographicAddress"
          },
          "city": "Paris",
          "country": "France",
          "postCode": "75014",
          "street1": "15 Rue des Canards"
        }
      ],
      "engagedParty": {
        "@type": "PartyRef",
        "href": "https://host:port/tmf-api/partyManagement/v4/organization/500",
        "id": "500",
        "name": "Happy Travellers",
        "@referredType": "Organization"
      },
      "name": "Moon Football Club",
      "relatedParty": [
        {
          "@type": "RelatedPartyOrPartyRole",
          "role": "bill receiver",
          "partyOrPartyRole": {
            "@type": "PartyRef",
```


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```
          "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
          "id": "2777",
          "@schemaLocation": "/schema-registry/schema/PartyRef",
          "@referredType": "Organization",
          "name": "John Doe (Accounting) Ltd"
        }
      }
    ],
    "status": "Approved",
    "validFor": {
      "startDateTime": "2018-06-12T00:00:00.000Z",
      "endDateTime": "2019-01-11T00:00:00.000Z"
    }
  },
  "account": [
    {
      "@type": "Customer360AccountVO",
      "href": "https://host:port/tmf-api/accountManagement/v5/account/5430",
      "id": "5430",
      "accountType": "Business",
      "description": "This billing account ...",
      "lastUpdate": "2018-06-14T10:30:00.000Z",
      "name": "Home Account",
      "state": "Inactive"
    }
  ],
  "agreement": [
    {
      "@type": "Customer360AgreementVO",
      "href": "https://host:port/tmf-api/agreementManagement/v4/agreement/28",
      "id": "28",
      "agreementPeriod": {
        "startDateTime": "2024-04-25T00:00:00.000Z",
        "endDateTime": "2028-04-25T00:00:00.000Z"
      },
      "agreementSpecification": {
        "@type": "AgreementSpecificationRef",
        "href": "https://host:port/tmf-
api/agreementManagement/v4/agreementSpecification/9834-11",
        "id": "9834-11",
        "name": "Bandwith extender",
        "@referredType": "AgreementSpecification",
        "description": "Special agreement with higher bandwith."
      },
      "agreementType": "commercial",
      "engagedParty": [
        {
          "@type": "PartyRef",
          "href": "https://host:port/tmf-api/partyManagement/v4/organization/1140",
          "id": "1140",
          "name": "Moon Football Club",
          "@referredType": "Organization"
        }
      ],
      "name": "Summer Contract Agreement"
    }
  ],
  "appointment": [
    {
      "@type": "Customer360AppointmentVO",
```


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```
      "href": "https://serverRoot/tmf-api/appointment/v4/appointment/21",
      "id": "21",
      "calendarEvent": {
        "@type": "CalendarEventRef",
        "href": "https://serverRoot/tmf-api/timeManagement/v4/calendarEvent/33",
        "id": "33",
        "name": "Intervention calendar event",
        "@referredType": "CalendarEvent"
      },
      "category": "intervention",
      "creationDate": "2018-02-01T14:40:43.000Z",
      "description": "Fix an internet connexion problem for a customer",
      "externalId": "432113",
      "lastUpdate": "2018-02-04T14:40:43.000Z",
      "status": "confirmed",
      "validFor": {
        "startDateTime": "2018-02-15T14:00:00.000Z",
        "endDateTime": "2018-02-15T16:00:00.000Z"
      }
    }
  ],
  "customerBill": [
    {
      "@type": "Customer360CustomerBillVO",
      "href": "https://server:port/tmf-api/Customer_Bill_Management/v5/customerBill/CB-
123",
      "id": "CB-123",
      "billNo": "33-34198237409810239487019873409123659128734879276359015ß2084370129347-
23A",
      "category": "Monthly",
      "state": "sent"
    },
    {
      "@type": "Customer360CustomerBillVO",
      "href": "https://server:port/tmf-api/Customer_Bill_Management/v5/customerBill/CB-
123",
      "id": "CB-123",
      "billNo": "780123456",
      "category": "normal",
      "state": "settled"
    }
  ],
  "loyaltyAccount": [
    {
      "@type": "Customer360LoyaltyAccountVO",
      "@baseType": "PartyAccount",
      "href": "https://host:port/tmf-api/loyaltyManagement/v5/loyaltyAccount/12345",
      "id": "12345",
      "description": "Loyalty account for loyalty program member",
      "lastUpdate": "2024-05-01T20:30:52.370Z",
      "name": "Loyalty Account",
      "state": "Active",
      "accountType": "Loyalty Account",
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Loyalty Program Member",
          "partyOrPartyRole": {
            "id": "1140",
            "href": "https://host:port/tmf-api/partyRoleManagement/v5/partyRole/1140",
```


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```
            "@type": "PartyRoleRef",
            "name": "Moon Football Club",
            "@referredType": "LoyaltyProgramMember"
          }
        }
      ],
      "accountBalance": [
        {
          "id": "AB123",
          "@type": "AccountBalance",
          "amount": {
            "value": 50,
            "unit": "POINTS"
          },
          "balanceType": "LoyaltyEarningsWallet",
          "validFor": {
            "startDateTime": "2021-03-17T00:00:00.000Z",
            "endDateTime": "2022-03-17T00:00:00.000Z"
          }
        }
      ]
    }
  ],
  "partyInteraction": [
    {
      "@type": "Customer360PartyInteractionVO",
      "href": "https://host:port/partyInteraction/v5/partyInteraction/123",
      "id": "123",
      "interactionDate": {
        "startDateTime": "2019-10-02T11:36:18.758Z",
        "endDateTime": "2019-10-02T11:53:21.789Z"
      },
      "description": "Visit to store",
      "reason": "The user wanted to buy TV and internet plan",
      "status": "ongoing"
    }
  ],
  "paymentMethod": [
    {
      "@type": "Customer360PaymentMethodVO",
      "href": "https://host:port/tmf-api/paymentMethod/v5/paymentMethod/12345",
      "id": "12345",
      "description": "My gold mastercard",
      "isPreferred": true,
      "validFor": {
        "startDateTime": "2021-03-17T00:00:00.000Z",
        "endDateTime": "2024-04-16T00:00:00.000Z"
      },
      "name": "Main credit card",
      "status": "active",
      "statusDate": "2021-03-17T00:00:00.000Z"
    }
  ],
  "productOrder": [
    {
      "@type": "Customer360ProductOrderVO",
      "href": "https://host:port/productOrderingManagement/v5/productOrder/30001",
      "id": "30001",
      "category": "B2C product order",
      "completionDate": "2019-05-02T08:13:59.506Z",
```


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```
      "description": "Product Order illustration sample",
      "expectedCompletionDate": "2019-05-02T08:13:59.506Z",
      "externalId": "PO-456",
      "creationDate": "2019-04-30T08:13:59.506Z",
      "priority": "1",
      "productOrderItem": [
        {
          "@type": "ProductOrderItem",
          "id": "1",
          "action": "add",
          "quantity": 1,
          "state": "completed"
        }
      ],
      "state": "completed"
    }
  ],
  "product": [
    {
      "@type": "Customer360ProductVO",
      "href": "https://host:port/productInventoryManagement/v5/product/g265-tf85",
      "id": "g265-tf85",
      "description": "product description",
      "name": "Voice Over IP Basic instance for Jean",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "href": "https://host:port/productCatalogManagement/v5/productOffering/PO-101-
1",
        "id": "PO-101-1",
        "name": "Voice Over IP Basic",
        "@referredType": "ProductOffering"
      },
      "status": "active"
    }
  ],
  "quote": [
    {
      "@type": "Customer360QuoteVO",
      "href": "https://host:port/quoteManagement/v4/quote/12dd-78hg",
      "id": "12dd-78hg",
      "category": "BSBS Quote",
      "creationDate": "2021-05-05T12:45:12.028Z",
      "description": "Quote illustration",
      "effectiveQuoteCompletionDate": "2021-05-21T12:45:12.028Z",
      "expectedFulfillmentStartDate": "2021-05-21T12:45:12.028Z",
      "expectedQuoteCompletionDate": "2021-05-14T12:45:12.028Z",
      "state": "approved",
      "validFor": {
        "startDateTime": "2021-05-06T12:45:12.033Z",
        "endDateTime": "2021-06-06T12:45:12.033Z"
      },
      "version": "1"
    }
  ],
  "serviceProblem": [
    {
      "@type": "Customer360ServiceProblemVO",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
      "id": "problemxxxx0000",
```


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```
      "affectedService": [
        {
          "@type": "ServiceRef",
          "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
          "id": "NW1_Tokyo_Osaka",
          "@referredType": "Service"
        },
        {
          "@type": "ServiceRef",
          "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
          "id": "NW1_Tokyo_xxxx",
          "@referredType": "Service"
        }
      ],
      "category": "supplier.originated",
      "description": "connection failure between Tokyo and Osaka",
      "originatingSystem": "System_001",
      "priority": 1,
      "reason": "Network infrastructure failure",
      "status": "resolved"
    }
  ],
  "troubleTicket": [
    {
      "@type": "Customer360TroubleTicketVO",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicket/v5/troubleTicket/3180",
      "id": "3180",
      "creationDate": "2019-05-31T07:34:45.968Z",
      "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle",
      "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
      "externalIdentifier": "213-9909",
      "lastUpdate": "2019-06-01T10:15:30.500Z",
      "name": "Compliant over last bill",
      "priority": "High",
      "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
      "severity": "Urgent",
      "status": "pending",
      "statusChangeDate": "2019-06-01T10:15:30.500Z",
      "statusChangeReason": "Need more information from the customer",
      "ticketType": "Bill Dispute"
    }
  ]
}
```


## Notification Resource Models

3 notifications are defined for this API.

Notifications related to Customer360:

- Create Event
- Delete Event

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- Attribute Value Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

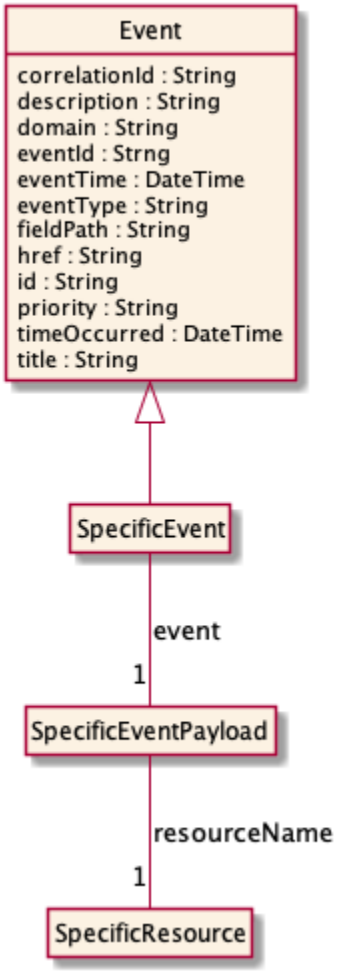



*Figure 21 Notification Pattern*

## Customer360

### Create Event

Message example for Customer360CreateEvent Notification

```
Content-Type: application/json
```

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```
{
  "@type": "Customer360CreateEvent",
  "@schemaLocation": "/schema-registry/schema/Customer360CreateEvent",
  "href": "/tmf-api/v5/Customer360CreateEvent/12345",
  "id": "12345",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "bill receiver",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
        "id": "2777",
        "@schemaLocation": "/schema-registry/schema/PartyRef",
        "@referredType": "Organization",
        "name": "John Doe (Accounting) Ltd"
      }
    }
  ],
  "eventId": "8a0a98df-2af3-4fbf-a54d-97f5944bc7cc",
  "eventTime": "2023-03-03T16:27:50.939Z",
  "@baseType": "Event",
  "eventType": "Customer360CreateEvent",
  "correlationId": "fe9b29c9-a31c",
  "domain": "Commercial",
  "title": "Customer360CreateEvent",
  "description": "Customer360CreateEvent illustration",
  "priority": "2",
  "timeOccurred": "2024-11-02T13:19:14.000Z",
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "schema-registry/schema/analyticCharacteristic",
      "id": "12345",
      "name": "ResponseTime",
      "valueType": "String",
      "value": "Analytic String",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "12345",
          "relationshipType": "internal"
        }
      ]
    }
  ],
  "source": {
    "@type": "ReportingResource",
    "id": "29",
    "name": "APP-171",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "ReportingResource",
    "id": "182",
    "name": "APP-116",
    "@referredType": "LogicalResource"
  },
  "event": {
```


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```
    "customer360": {
      "@type": "Customer360",
      "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
      "id": "1140",
      "@schemaLocation": "/schema-registry/schema/Customer360",
      "validFor": {
        "startDateTime": "2018-06-13T00:00:00.000Z",
        "endDateTime": "2019-01-11T00:00:00.000Z"
      },
      "customer": {
        "@type": "Customer360CustomerVO",
        "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
        "id": "1140",
        "contactMedium": [
          {
            "@type": "PhoneContactMedium",
            "@baseType": "ContactMedium",
            "@schemaLocation": "/schema-registry/schema/PhoneContactMedium",
            "id": "2256789876",
            "preferred": true,
            "contactType": "homePhone",
            "validFor": {
              "startDateTime": "2018-06-13T00:00:00.000Z",
              "endDateTime": "2019-01-11T00:00:00.000Z"
            },
            "phoneNumber": "01 09 75 83 51"
          },
          {
            "@type": "GeographicAddressContactMedium",
            "@baseType": "ContactMedium",
            "@schemaLocation": "/schema-
registry/schema/GeographicAddressContactMedium",
            "id": "12345",
            "stateOrProvince": "NY",
            "street2": "123 Lindsay Street",
            "preferred": false,
            "contactType": "homeAddress",
            "validFor": {
              "startDateTime": "2018-06-13T00:00:00.000Z",
              "endDateTime": "2019-01-11T00:00:00.000Z"
            },
            "city": "Paris",
            "geographicAddress": {
              "@type": "GeographicAddressRef",
              "href": "https://host:port/tmf-
api/geographicAddressManagement/v5/geographicAddress/12345",
              "id": "12345",
              "@referredType": "GeographicAddress"
            },
            "country": "France",
            "postCode": "75014",
            "street1": "15 Rue des Canards"
          }
        ],
        "engagedParty": {
          "@type": "PartyRef",
          "href": "https://host:port/tmf-api/partyManagement/v4/organization/500",
          "id": "500",
          "name": "Happy Travellers",
          "@referredType": "Organization"
```


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```
        },
        "name": "Moon Football Club",
        "relatedParty": [
          {
            "@type": "RelatedPartyOrPartyRole",
            "role": "bill receiver",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
              "id": "2777",
              "@schemaLocation": "/schema-registry/schema/PartyRef",
              "@referredType": "Organization",
              "name": "John Doe (Accounting) Ltd"
            }
          }
        ],
        "status": "Approved",
        "validFor": {
          "startDateTime": "2018-06-12T00:00:00.000Z",
          "endDateTime": "2019-01-11T00:00:00.000Z"
        }
      },
      "account": [
        {
          "@type": "Customer360AccountVO",
          "href": "https://host:port/tmf-api/accountManagement/v5/account/5430",
          "id": "5430",
          "accountType": "Business",
          "description": "This billing account ...",
          "lastUpdate": "2018-06-14T10:30:00.000Z",
          "name": "Home Account",
          "state": "Inactive"
        }
      ],
      "agreement": [
        {
          "@type": "Customer360AgreementVO",
          "href": "https://host:port/tmf-api/agreementManagement/v4/agreement/28",
          "id": "28",
          "agreementPeriod": {
            "startDateTime": "2024-04-25T00:00:00.000Z",
            "endDateTime": "2028-04-25T00:00:00.000Z"
          },
          "agreementSpecification": {
            "@type": "AgreementSpecificationRef",
            "href": "https://host:port/tmf-
api/agreementManagement/v4/agreementSpecification/9834-11",
            "id": "9834-11",
            "name": "Bandwith extender",
            "@referredType": "AgreementSpecification",
            "description": "Special agreement with higher bandwith."
          },
          "agreementType": "commercial",
          "name": "Summer Contract Agreement"
        }
      ],
      "appointment": [
        {
          "@type": "Customer360AppointmentVO",
          "href": "https://serverRoot/tmf-api/appointment/v4/appointment/21",
```


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```
          "id": "21",
          "calendarEvent": {
            "@type": "CalendarEventRef",
            "href": "https://serverRoot/tmf-api/timeManagement/v4/calendarEvent/33",
            "id": "33",
            "name": "Intervention calendar event",
            "@referredType": "CalendarEvent"
          },
          "category": "intervention",
          "creationDate": "2018-02-01T14:40:43.000Z",
          "description": "Fix an internet connexion problem for a customer",
          "externalId": "432113",
          "lastUpdate": "2018-02-04T14:40:43.000Z",
          "status": "confirmed",
          "validFor": {
            "startDateTime": "2018-02-15T14:00:00.000Z",
            "endDateTime": "2018-02-15T16:00:00.000Z"
          }
        }
      ],
      "customerBill": [
        {
          "@type": "Customer360CustomerBillVO",
          "href": "https://server:port/tmf-
api/Customer_Bill_Management/v5/customerBill/CB-123",
          "id": "CB-123",
          "billNo": "33-
341982374098102394870198734091236591287348792763590lß2084370129347-23A",
          "category": "Monthly",
          "state": "sent"
        },
        {
          "@type": "Customer360CustomerBillVO",
          "href": "https://server:port/tmf-
api/Customer_Bill_Management/v5/customerBill/CB-123",
          "id": "CB-123",
          "billNo": "780123456",
          "category": "normal",
          "state": "settled"
        }
      ],
      "loyaltyAccount": [
        {
          "@type": "Customer360LoyaltyAccountVO",
          "@baseType": "PartyAccount",
          "href": "https://host:port/tmf-
api/loyaltyManagement/v5/loyaltyAccount/12345",
          "id": "12345",
          "description": "Loyalty account for loyalty program member",
          "lastUpdate": "2024-05-01T20:30:52.370Z",
          "name": "Loyalty Account",
          "state": "Active",
          "accountType": "Loyalty Account",
          "relatedParty": [
            {
              "@type": "RelatedPartyRefOrPartyRoleRef",
              "role": "Loyalty Program Member",
              "partyOrPartyRole": {
                "id": "1140",
                "href": "https://host:port/tmf-
```


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```
api/partyRoleManagement/v5/partyRole/1140",
                "@type": "PartyRoleRef",
                "name": "Moon Football Club",
                "@referredType": "LoyaltyProgramMember"
              }
            }
          ],
          "accountBalance": [
            {
              "id": "AB123",
              "@type": "AccountBalance",
              "amount": {
                "value": 50,
                "unit": "POINTS"
              },
              "balanceType": "LoyaltyEarningsWallet",
              "validFor": {
                "startDateTime": "2021-03-17T00:00:00.000Z",
                "endDateTime": "2022-03-17T00:00:00.000Z"
              }
            }
          ]
        }
      ],
      "partyInteraction": [
        {
          "@type": "Customer360PartyInteractionVO",
          "href": "https://host:port/partyInteraction/v5/partyInteraction/123",
          "id": "123",
          "interactionDate": {
            "startDateTime": "2019-10-02T11:36:18.758Z",
            "endDateTime": "2019-10-02T11:53:21.789Z"
          },
          "description": "Visit to store",
          "reason": "The user wanted to buy TV and internet plan",
          "status": "ongoing"
        }
      ],
      "paymentMethod": [
        {
          "@type": "Customer360PaymentMethodVO",
          "href": "https://host:port/tmf-api/paymentMethod/v5/paymentMethod/12345",
          "id": "12345",
          "description": "My gold mastercard",
          "isPreferred": true,
          "status": "Active",
          "statusDate": "2024-04-16T00:00:00.000Z",
          "validFor": {
            "startDateTime": "2021-03-17T00:00:00.000Z",
            "endDateTime": "2024-04-16T00:00:00.000Z"
          },
          "name": "Main credit card"
        }
      ],
      "productOrder": [
        {
          "@type": "Customer360ProductOrderVO",
          "href": "https://host:port/productOrderingManagement/v5/productOrder/30001",
          "id": "30001",
          "category": "B2C product order",
```


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```
          "completionDate": "2019-05-02T08:13:59.506Z",
          "description": "Product Order illustration sample",
          "expectedCompletionDate": "2019-05-02T08:13:59.506Z",
          "externalId": "PO-456",
          "creationDate": "2019-04-30T08:13:59.506Z",
          "priority": "1",
          "state": "completed"
        }
      ],
      "product": [
        {
          "@type": "Customer360ProductVO",
          "href": "https://host:port/productInventoryManagement/v5/product/g265-tf85",
          "id": "g265-tf85",
          "description": "product description",
          "name": "Voice Over IP Basic instance for Jean",
          "productOffering": {
            "@type": "ProductOfferingRef",
            "href": "https://host:port/productCatalogManagement/v5/productOffering/PO-
101-1",
            "id": "PO-101-1",
            "name": "Voice Over IP Basic",
            "@referredType": "ProductOffering"
          },
          "status": "active"
        }
      ],
      "quote": [
        {
          "@type": "Customer360QuoteVO",
          "href": "https://host:port/quoteManagement/v4/quote/12dd-78hg",
          "id": "12dd-78hg",
          "category": "BSBS Quote",
          "description": "Quote illustration",
          "effectiveQuoteCompletionDate": "2021-05-21T12:45:12.028Z",
          "expectedFulfillmentStartDate": "2021-05-21T12:45:12.028Z",
          "expectedQuoteCompletionDate": "2021-05-14T12:45:12.028Z",
          "state": "approved",
          "validFor": {
            "startDateTime": "2021-05-06T12:45:12.033Z",
            "endDateTime": "2021-06-06T12:45:12.033Z"
          },
          "version": "1"
        }
      ],
      "serviceProblem": [
        {
          "@type": "Customer360ServiceProblemVO",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
          "id": "problemxxxx0000",
          "affectedService": [
            {
              "@type": "ServiceRef",
              "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
              "id": "NW1_Tokyo_Osaka",
              "@referredType": "Service"
            },
            {
```


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```
              "@type": "ServiceRef",
              "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
              "id": "NW1_Tokyo_xxxx",
              "@referredType": "Service"
            }
          ],
          "category": "supplier.originated",
          "description": "connection failure between Tokyo and Osaka",
          "originatingSystem": "System_001",
          "priority": 1,
          "status": "resolved"
        }
      ],
      "troubleTicket": [
        {
          "@type": "Customer360TroubleTicketVO",
          "href": "https://mycsp.com:8080/tmf-api/troubleTicket/v5/troubleTicket/3180",
          "id": "3180",
          "creationDate": "2019-05-31T07:34:45.968Z",
          "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle",
          "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
          "externalIdentifier": "213-9909",
          "lastUpdate": "2019-06-01T10:15:30.500Z",
          "name": "Compliant over last bill",
          "priority": "High",
          "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
          "severity": "Urgent",
          "status": "pending",
          "statusChangeDate": "2019-06-01T10:15:30.500Z",
          "statusChangeReason": "Need more information from the customer",
          "ticketType": "Bill Dispute"
        }
      ]
    }
  }
}
```


### Delete Event

Message example for Customer360DeleteEvent Notification


```
Content-Type: application/json
{
  "event": {
    "customer360": {
      "@type": "Customer360",
      "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
      "id": "1140",
      "@schemaLocation": "/schema-registry/schema/Customer360",
      "customer": {
        "@type": "Customer360CustomerVO",
        "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
        "id": "1140",
        "name": "Moon Football Club",
```


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```
        "status": "Approved"
      }
    }
  },
  "@type": "Customer360DeleteEvent",
  "@baseType": "Event",
  "@schemaLocation": "/schema-registry/schema/customer360DeleteEvent",
  "href": "/tmf-api/v5/customer360DeleteEvent/12345",
  "id": "12345",
  "eventId": "8a0a98df-2af3-4fbf-a54d-97f5944bc7cc",
  "eventTime": "2023-03-03T16:27:50.939Z",
  "eventType": "Customer360DeleteEvent",
  "correlationId": "fe9b29c9-a31c",
  "domain": "Commercial",
  "title": "Customer360DeleteEvent",
  "description": "Customer360DeleteEvent illustration",
  "priority": "9",
  "timeOccurred": "2024-11-02T13:19:14.000Z",
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "schema-registry/schema/analyticCharacteristic",
      "id": "12345",
      "name": "ResponseTime",
      "valueType": "String",
      "value": "Analytic String",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "12345",
          "relationshipType": "internal"
        }
      ]
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "bill receiver",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
        "id": "2777",
        "@schemaLocation": "/schema-registry/schema/PartyRef",
        "@referredType": "Organization",
        "name": "John Doe (Accounting) Ltd"
      }
    }
  ],
  "source": {
    "@type": "ReportingResource",
    "id": "29",
    "name": "APP-19981",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "ReportingResource",
    "id": "182",
    "name": "APP-19981",
```


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```
    "@referredType": "LogicalResource"
  }
}
```

### Attribute Value Change Event

Message example for Customer360AttributeValueChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "Customer360AttributeValueChangeEvent",
  "@baseType": "Event",
  "@schemaLocation": "/schema-registry/schema/Customer360AttributeValueChangeEvent",
  "href": "/tmf-api/v5/Customer360AttributeValueChangeEvent/12345",
  "id": "12345",
  "eventId": "8a0a98df-2af3-4fbf-a54f-97f5944bc7cc",
  "eventTime": "2023-03-03T16:27:50.939Z",
  "eventType": "Customer360AttributeValueChangeEvent",
  "correlationId": "fe9b29c9-a31c",
  "domain": "Commercial",
  "title": "Customer360AttributeValueChangeEvent",
  "description": "Customer360AttributeValueChangeEvent illustration",
  "priority": "2",
  "analyticCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "schema-registry/schema/analyticCharacteristic",
      "id": "12345",
      "name": "ResponseTime",
      "valueType": "String",
      "value": "Analytic String",
      "characteristicRelationship": [
        {
          "@type": "CharacteristicRelationship",
          "id": "12345",
          "relationshipType": "internal"
        }
      ]
    }
  ],
  "timeOccurred": "2024-11-02T13:19:14.000Z",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "bill receiver",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
        "id": "2777",
        "@schemaLocation": "/schema-registry/schema/PartyRef",
        "@referredType": "Organization",
        "name": "John Doe (Accounting) Ltd"
      }
    }
  ],
```

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```
  "source": {
    "@type": "ReportingResource",
    "id": "29",
    "name": "APP-171",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "ReportingResource",
    "id": "182",
    "name": "APP-116",
    "@referredType": "LogicalResource"
  },
  "event": {
    "customer360": {
      "@type": "Customer360",
      "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
      "id": "1140",
      "@schemaLocation": "/schema-registry/schema/Customer360",
      "validFor": {
        "startDateTime": "2018-06-13T00:00:00.000Z",
        "endDateTime": "2019-06-11T00:00:00.000Z"
      },
      "customer": {
        "@type": "Customer360CustomerVO",
        "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
        "id": "1140",
        "name": "Moon Football Club",
        "status": "Approved"
      }
    }
  }
}
```


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# API OPERATIONS

Remember the following Uniform Contract:

| Operation on Entities | Uniform API Operation | Description |
|---|---|---|
| Query Entities | GET Resource | GET must be used to retrieve a representation of a resource. |
| Create Entity | POST Resource | POST must be used to create a new resource |
| Partial Update of an Entity | PATCH Resource | PATCH must be used to partially update a resource |
| Remove an Entity | DELETE Resource | DELETE must be used to remove a resource |
| Execute an Action on an Entity | POST on TASK Resource | POST must be used to execute Task Resources |
| Other Request Methods | POST on TASK Resource | GET and POST must not be used to tunnel other request methods. |

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

## Operations on Customer360

### Retrieves a customer360 by ID

`GET /customer360/{id}?fields=…`

**Description**

This operation retrieves a Customer360 entity. Attribute selection is enabled for all first level attributes.

**Usage samples**

Customer360 retrieve example

**Request**

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```
GET /customer360/1142
Content-Type: application/json
```

### Response

```
200

{
  "@type": "Customer360",
  "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
  "id": "1140",
  "@schemaLocation": "/schema-registry/schema/Customer360",
  "validFor": {
    "startDateTime": "2018-06-13T00:00:00.000Z",
    "endDateTime": "2019-01-11T00:00:00.000Z"
  },
  "customer": {
    "@type": "Customer360CustomerVO",
    "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
    "id": "1140",
    "contactMedium": [
      {
        "@type": "PhoneContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/schema/PhoneContactMedium",
        "id": "2256789876",
        "preferred": true,
        "contactType": "homePhone",
        "validFor": {
          "startDateTime": "2018-06-13T00:00:00.000Z",
          "endDateTime": "2019-01-11T00:00:00.000Z"
        },
        "phoneNumber": "01 09 75 83 51"
      },
      {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/schema/GeographicAddressContactMedium",
        "id": "12345",
        "stateOrProvince": "NY",
        "street2": "123 Lindsay Street",
        "preferred": false,
        "contactType": "homeAddress",
        "validFor": {
          "startDateTime": "2018-06-13T00:00:00.000Z",
          "endDateTime": "2019-01-11T00:00:00.000Z"
        },
        "geographicAddress": {
          "@type": "GeographicAddressRef",
          "href": "https://host:port/tmf-
api/geographicAddressManagement/v5/geographicAddress/12345",
          "id": "12345",
          "@referredType": "GeographicAddress"
        },
        "city": "Paris",
        "country": "France",
```

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```
        "postCode": "75014",
        "street1": "15 Rue des Canards"
      }
    ],
    "engagedParty": {
      "@type": "PartyRef",
      "href": "https://host:port/tmf-api/partyManagement/v4/organization/500",
      "id": "500",
      "name": "Happy Travellers",
      "@referredType": "Organization"
    },
    "name": "Moon Football Club",
    "relatedParty": [
      {
        "@type": "RelatedPartyOrPartyRole",
        "role": "bill receiver",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
          "id": "2777",
          "@schemaLocation": "/schema-registry/schema/PartyRef",
          "@referredType": "Organization",
          "name": "John Doe (Accounting) Ltd"
        }
      }
    ],
    "status": "Approved",
    "validFor": {
      "startDateTime": "2018-06-12T00:00:00.000Z",
      "endDateTime": "2019-01-11T00:00:00.000Z"
    }
  },
  "account": [
    {
      "@type": "Customer360AccountVO",
      "href": "https://host:port/tmf-api/accountManagement/v5/account/5430",
      "id": "5430",
      "accountType": "Business",
      "description": "This billing account ...",
      "lastUpdate": "2018-06-14T10:30:00.000Z",
      "name": "Home Account",
      "state": "Inactive"
    }
  ],
  "agreement": [
    {
      "@type": "Customer360AgreementVO",
      "href": "https://host:port/tmf-api/agreementManagement/v4/agreement/28",
      "id": "28",
      "agreementPeriod": {
        "startDateTime": "2024-04-25T00:00:00.000Z",
        "endDateTime": "2028-04-25T00:00:00.000Z"
      },
      "agreementSpecification": {
        "@type": "AgreementSpecificationRef",
        "href": "https://host:port/tmf-
api/agreementManagement/v4/agreementSpecification/9834-11",
        "id": "9834-11",
        "name": "Bandwith extender",
        "@referredType": "AgreementSpecification",
```


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```
        "description": "Special agreement with higher bandwith."
      },
      "agreementType": "commercial",
      "engagedParty": [
        {
          "@type": "PartyRef",
          "href": "https://host:port/tmf-api/partyManagement/v4/organization/1140",
          "id": "1140",
          "name": "Moon Football Club",
          "@referredType": "Organization"
        }
      ],
      "name": "Summer Contract Agreement"
    }
  ],
  "appointment": [
    {
      "@type": "Customer360AppointmentVO",
      "href": "https://serverRoot/tmf-api/appointment/v4/appointment/21",
      "id": "21",
      "calendarEvent": {
        "@type": "CalendarEventRef",
        "href": "https://serverRoot/tmf-api/timeManagement/v4/calendarEvent/33",
        "id": "33",
        "name": "Intervention calendar event",
        "@referredType": "CalendarEvent"
      },
      "category": "intervention",
      "creationDate": "2018-02-01T14:40:43.000Z",
      "description": "Fix an internet connexion problem for a customer",
      "externalId": "432113",
      "lastUpdate": "2018-02-04T14:40:43.000Z",
      "status": "confirmed",
      "validFor": {
        "startDateTime": "2018-02-15T14:00:00.000Z",
        "endDateTime": "2018-02-15T16:00:00.000Z"
      }
    }
  ],
  "customerBill": [
    {
      "@type": "Customer360CustomerBillVO",
      "href": "https://server:port/tmf-api/Customer_Bill_Management/v5/customerBill/CB-
123",
      "id": "CB-123",
      "billNo": "33-34198237409810239487019873409123659128734879276359O1ß2084370129347-
23A",
      "category": "Monthly",
      "state": "sent"
    },
    {
      "@type": "Customer360CustomerBillVO",
      "href": "https://server:port/tmf-api/Customer_Bill_Management/v5/customerBill/CB-
123",
      "id": "CB-123",
      "billNo": "780123456",
      "category": "normal",
      "state": "settled"
    }
  ],
```


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```
  "loyaltyAccount": [
    {
      "@type": "Customer360LoyaltyAccountVO",
      "@baseType": "PartyAccount",
      "href": "https://host:port/tmf-api/loyaltyManagement/v5/loyaltyAccount/12345",
      "id": "12345",
      "description": "Loyalty account for loyalty program member",
      "lastUpdate": "2024-05-01T20:30:52.370Z",
      "name": "Loyalty Account",
      "state": "Active",
      "accountType": "Loyalty Account",
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Loyalty Program Member",
          "partyOrPartyRole": {
            "id": "1140",
            "href": "https://host:port/tmf-api/partyRoleManagement/v5/partyRole/1140",
            "@type": "PartyRoleRef",
            "name": "Moon Football Club",
            "@referredType": "LoyaltyProgramMember"
          }
        }
      ],
      "accountBalance": [
        {
          "id": "AB123",
          "@type": "AccountBalance",
          "amount": {
            "value": 50,
            "unit": "POINTS"
          },
          "balanceType": "LoyaltyEarningsWallet",
          "validFor": {
            "startDateTime": "2021-03-17T00:00:00.000Z",
            "endDateTime": "2022-03-17T00:00:00.000Z"
          }
        }
      ]
    }
  ],
  "partyInteraction": [
    {
      "@type": "Customer360PartyInteractionVO",
      "href": "https://host:port/partyInteraction/v5/partyInteraction/123",
      "id": "123",
      "interactionDate": {
        "startDateTime": "2019-10-02T11:36:18.758Z",
        "endDateTime": "2019-10-02T11:53:21.789Z"
      },
      "description": "Visit to store",
      "reason": "The user wanted to buy TV and internet plan",
      "status": "ongoing"
    }
  ],
  "paymentMethod": [
    {
      "@type": "Customer360PaymentMethodVO",
      "href": "https://host:port/tmf-api/paymentMethod/v5/paymentMethod/12345",
      "id": "12345",
```


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```
      "description": "My gold mastercard",
      "isPreferred": true,
      "validFor": {
        "startDateTime": "2021-03-17T00:00:00.000Z",
        "endDateTime": "2024-04-16T00:00:00.000Z"
      },
      "name": "Main credit card",
      "status": "active",
      "statusDate": "2021-03-17T00:00:00.000Z"
    }
  ],
  "productOrder": [
    {
      "@type": "Customer360ProductOrderVO",
      "href": "https://host:port/productOrderingManagement/v5/productOrder/30001",
      "id": "30001",
      "category": "B2C product order",
      "completionDate": "2019-05-02T08:13:59.506Z",
      "description": "Product Order illustration sample",
      "expectedCompletionDate": "2019-05-02T08:13:59.506Z",
      "externalId": "PO-456",
      "creationDate": "2019-04-30T08:13:59.506Z",
      "priority": "1",
      "productOrderItem": [
        {
          "@type": "ProductOrderItem",
          "id": "1",
          "action": "add",
          "quantity": 1,
          "state": "completed"
        }
      ],
      "state": "completed"
    }
  ],
  "product": [
    {
      "@type": "Customer360ProductVO",
      "href": "https://host:port/productInventoryManagement/v5/product/g265-tf85",
      "id": "g265-tf85",
      "description": "product description",
      "name": "Voice Over IP Basic instance for Jean",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "href": "https://host:port/productCatalogManagement/v5/productOffering/PO-101-
1",
        "id": "PO-101-1",
        "name": "Voice Over IP Basic",
        "@referredType": "ProductOffering"
      },
      "status": "active"
    }
  ],
  "quote": [
    {
      "@type": "Customer360QuoteVO",
      "href": "https://host:port/quoteManagement/v4/quote/12dd-78hg",
      "id": "12dd-78hg",
      "category": "BSBS Quote",
      "creationDate": "2021-05-05T12:45:12.028Z",
```


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```
      "description": "Quote illustration",
      "effectiveQuoteCompletionDate": "2021-05-21T12:45:12.028Z",
      "expectedFulfillmentStartDate": "2021-05-21T12:45:12.028Z",
      "expectedQuoteCompletionDate": "2021-05-14T12:45:12.028Z",
      "state": "approved",
      "validFor": {
        "startDateTime": "2021-05-06T12:45:12.033Z",
        "endDateTime": "2021-06-06T12:45:12.033Z"
      },
      "version": "1"
    }
  ],
  "serviceProblem": [
    {
      "@type": "Customer360ServiceProblemVO",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
      "id": "problemxxxx0000",
      "affectedService": [
        {
          "@type": "ServiceRef",
          "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
          "id": "NW1_Tokyo_Osaka",
          "@referredType": "Service"
        },
        {
          "@type": "ServiceRef",
          "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
          "id": "NW1_Tokyo_xxxx",
          "@referredType": "Service"
        }
      ],
      "category": "supplier.originated",
      "description": "connection failure between Tokyo and Osaka",
      "originatingSystem": "System_001",
      "priority": 1,
      "reason": "Network infrastructure failure",
      "status": "resolved"
    }
  ],
  "troubleTicket": [
    {
      "@type": "Customer360TroubleTicketVO",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicket/v5/troubleTicket/3180",
      "id": "3180",
      "creationDate": "2019-05-31T07:34:45.968Z",
      "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle",
      "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
      "externalIdentifier": "213-9909",
      "lastUpdate": "2019-06-01T10:15:30.500Z",
      "name": "Compliant over last bill",
      "priority": "High",
      "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
      "severity": "Urgent",
      "status": "pending",
      "statusChangeDate": "2019-06-01T10:15:30.500Z",
      "statusChangeReason": "Need more information from the customer",
```


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```
      "ticketType": "Bill Dispute"
    }
  ]
}
```

## List or find customer360 objects

### `GET /customer360?fields=…&{filtering}`

#### Description

This operation list Customer360 entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Customer360 list example

#### Request

```
GET /customer360
Content-Type: application/json
```

#### Response

```
200

[
  {
    "@type": "Customer360",
    "href": "https://host:port/tmf-api/customer360/v5/customer360/1140",
    "id": "1140",
    "@schemaLocation": "/schema-registry/schema/Customer360",
    "validFor": {
      "startDateTime": "2018-06-13T00:00:00.000Z",
      "endDateTime": "2019-01-11T00:00:00.000Z"
    },
    "customer": {
      "@type": "Customer360CustomerVO",
      "href": "https://host:port/tmf-api/customerManagement/v5/customer/1140",
      "id": "1140",
      "contactMedium": [
        {
          "@type": "PhoneContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/schema/PhoneContactMedium",
          "id": "2256789876",
          "preferred": true,
          "contactType": "homePhone",
          "validFor": {
```

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```
            "startDateTime": "2018-06-13T00:00:00.000Z",
            "endDateTime": "2019-01-11T00:00:00.000Z"
          },
          "phoneNumber": "01 09 75 83 51"
        },
        {
          "@type": "GeographicAddressContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/schema/GeographicAddressContactMedium",
          "id": "12345",
          "stateOrProvince": "NY",
          "street2": "123 Lindsay Street",
          "preferred": false,
          "contactType": "homeAddress",
          "validFor": {
            "startDateTime": "2018-06-13T00:00:00.000Z",
            "endDateTime": "2019-01-11T00:00:00.000Z"
          },
          "city": "Paris",
          "country": "France",
          "postCode": "75014",
          "geographicAddress": {
            "@type": "GeographicAddressRef",
            "href": "https://host:port/tmf-
api/geographicAddressManagement/v5/geographicAddress/12345",
            "id": "12345",
            "@referredType": "GeographicAddress"
          },
          "street1": "15 Rue des Canards"
        }
      ],
      "engagedParty": {
        "@type": "PartyRef",
        "href": "https://host:port/tmf-api/partyManagement/v4/organization/500",
        "id": "500",
        "name": "Happy Travellers",
        "@referredType": "Organization"
      },
      "name": "Moon Football Club",
      "relatedParty": [
        {
          "@type": "RelatedPartyOrPartyRole",
          "role": "bill receiver",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "https://host:port/tmf-api/partyManagement/v5/organization/2777",
            "id": "2777",
            "@schemaLocation": "/schema-registry/schema/PartyRef",
            "@referredType": "Organization",
            "name": "John Doe (Accounting) Ltd"
          }
        }
      ],
      "status": "Approved",
      "validFor": {
        "startDateTime": "2018-06-12T00:00:00.000Z",
        "endDateTime": "2019-01-11T00:00:00.000Z"
      }
    },
    "account": [
```


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```
      {
        "@type": "Customer360AccountVO",
        "href": "https://host:port/tmf-api/accountManagement/v5/account/5430",
        "id": "5430",
        "accountType": "Business",
        "description": "This billing account ...",
        "lastUpdate": "2018-06-14T10:30:00.000Z",
        "name": "Home Account",
        "state": "Inactive"
      }
    ],
    "agreement": [
      {
        "@type": "Customer360AgreementVO",
        "href": "https://host:port/tmf-api/agreementManagement/v4/agreement/28",
        "id": "28",
        "agreementPeriod": {
          "startDateTime": "2024-04-25T00:00:00.000Z",
          "endDateTime": "2028-04-25T00:00:00.000Z"
        },
        "agreementSpecification": {
          "@type": "AgreementSpecificationRef",
          "href": "https://host:port/tmf-
api/agreementManagement/v4/agreementSpecification/9834-11",
          "id": "9834-11",
          "name": "Bandwith extender",
          "@referredType": "AgreementSpecification",
          "description": "Special agreement with higher bandwith."
        },
        "agreementType": "commercial",
        "engagedParty": [
          {
            "@type": "PartyRef",
            "href": "https://host:port/tmf-api/partyManagement/v4/organization/1140",
            "id": "1140",
            "name": "Moon Football Club",
            "@referredType": "Organization"
          }
        ],
        "name": "Summer Contract Agreement"
      }
    ],
    "appointment": [
      {
        "@type": "Customer360AppointmentVO",
        "href": "https://serverRoot/tmf-api/appointment/v4/appointment/21",
        "id": "21",
        "calendarEvent": {
          "@type": "CalendarEventRef",
          "href": "https://serverRoot/tmf-api/timeManagement/v4/calendarEvent/33",
          "id": "33",
          "name": "Intervention calendar event",
          "@referredType": "CalendarEvent"
        },
        "category": "intervention",
        "creationDate": "2018-02-01T14:40:43.000Z",
        "description": "Fix an internet connexion problem for a customer",
        "externalId": "432113",
        "lastUpdate": "2018-02-04T14:40:43.000Z",
        "status": "confirmed",
```


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```
        "validFor": {
          "startDateTime": "2018-02-15T14:00:00.000Z",
          "endDateTime": "2018-02-15T16:00:00.000Z"
        }
      }
    ],
    "customerBill": [
      {
        "@type": "Customer360CustomerBillVO",
        "href": "https://server:port/tmf-
api/Customer_Bill_Management/v5/customerBill/CB-123",
        "id": "CB-123",
        "billNo": "33-
341982374098102394870198734091236591287348792763590lß2084370129347-23A",
        "category": "Monthly",
        "state": "sent"
      },
      {
        "@type": "Customer360CustomerBillVO",
        "href": "https://server:port/tmf-
api/Customer_Bill_Management/v5/customerBill/CB-123",
        "id": "CB-123",
        "billNo": "780123456",
        "category": "normal",
        "state": "settled"
      }
    ],
    "loyaltyAccount": [
      {
        "@type": "Customer360LoyaltyAccountVO",
        "@baseType": "PartyAccount",
        "href": "https://host:port/tmf-api/loyaltyManagement/v5/loyaltyAccount/12345",
        "id": "12345",
        "description": "Loyalty account for loyalty program member",
        "lastUpdate": "2024-05-01T20:30:52.370Z",
        "name": "Loyalty Account",
        "state": "Active",
        "accountType": "Loyalty Account",
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Loyalty Program Member",
            "partyOrPartyRole": {
              "id": "1140",
              "href": "https://host:port/tmf-
api/partyRoleManagement/v5/partyRole/1140",
              "@type": "PartyRoleRef",
              "name": "Moon Football Club",
              "@referredType": "LoyaltyProgramMember"
            }
          }
        ],
        "accountBalance": [
          {
            "id": "AB123",
            "@type": "AccountBalance",
            "amount": {
              "value": 50,
              "unit": "POINTS"
            },
```


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```
            "balanceType": "LoyaltyEarningsWallet",
            "validFor": {
              "startDateTime": "2021-03-17T00:00:00.000Z",
              "endDateTime": "2022-03-17T00:00:00.000Z"
            }
          }
        ]
      }
    ],
    "partyInteraction": [
      {
        "@type": "Customer360PartyInteractionVO",
        "href": "https://host:port/partyInteraction/v5/partyInteraction/123",
        "id": "123",
        "interactionDate": {
          "startDateTime": "2019-10-02T11:36:18.758Z",
          "endDateTime": "2019-10-02T11:53:21.789Z"
        },
        "description": "Visit to store",
        "reason": "The user wanted to buy TV and internet plan",
        "status": "ongoing"
      }
    ],
    "paymentMethod": [
      {
        "@type": "Customer360PaymentMethodVO",
        "href": "https://host:port/tmf-api/paymentMethod/v5/paymentMethod/12345",
        "id": "12345",
        "description": "My gold mastercard",
        "isPreferred": true,
        "validFor": {
          "startDateTime": "2021-03-17T00:00:00.000Z",
          "endDateTime": "2024-04-16T00:00:00.000Z"
        },
        "name": "Main credit card",
        "status": "active",
        "statusDate": "2021-03-17T00:00:00.000Z"
      }
    ],
    "productOrder": [
      {
        "@type": "Customer360ProductOrderVO",
        "href": "https://host:port/productOrderingManagement/v5/productOrder/30001",
        "id": "30001",
        "category": "B2C product order",
        "completionDate": "2019-05-02T08:13:59.506Z",
        "description": "Product Order illustration sample",
        "expectedCompletionDate": "2019-05-02T08:13:59.506Z",
        "externalId": "PO-456",
        "creationDate": "2019-04-30T08:13:59.506Z",
        "priority": "1",
        "productOrderItem": [
          {
            "@type": "ProductOrderItem",
            "id": "1",
            "action": "add",
            "quantity": 1,
            "state": "completed"
          }
        ],
```


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```
        "state": "completed"
      }
    ],
    "product": [
      {
        "@type": "Customer360ProductVO",
        "href": "https://host:port/productInventoryManagement/v5/product/g265-tf85",
        "id": "g265-tf85",
        "description": "product description",
        "name": "Voice Over IP Basic instance for Jean",
        "productOffering": {
          "@type": "ProductOfferingRef",
          "href": "https://host:port/productCatalogManagement/v5/productOffering/PO-
101-1",
          "id": "PO-101-1",
          "name": "Voice Over IP Basic",
          "@referredType": "ProductOffering"
        },
        "status": "active"
      }
    ],
    "quote": [
      {
        "@type": "Customer360QuoteVO",
        "href": "https://host:port/quoteManagement/v4/quote/12dd-78hg",
        "id": "12dd-78hg",
        "category": "BSBS Quote",
        "creationDate": "2021-05-05T12:45:12.028Z",
        "description": "Quote illustration",
        "effectiveQuoteCompletionDate": "2021-05-21T12:45:12.028Z",
        "expectedFulfillmentStartDate": "2021-05-21T12:45:12.028Z",
        "expectedQuoteCompletionDate": "2021-05-14T12:45:12.028Z",
        "state": "approved",
        "validFor": {
          "startDateTime": "2021-05-06T12:45:12.033Z",
          "endDateTime": "2021-06-06T12:45:12.033Z"
        },
        "version": "1"
      }
    ],
    "serviceProblem": [
      {
        "@type": "Customer360ServiceProblemVO",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
        "id": "problemxxxx0000",
        "affectedService": [
          {
            "@type": "ServiceRef",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
            "id": "NW1_Tokyo_Osaka",
            "@referredType": "Service"
          },
          {
            "@type": "ServiceRef",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
            "id": "NW1_Tokyo_xxxx",
            "@referredType": "Service"
```


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```
          }
        ],
        "category": "supplier.originated",
        "description": "connection failure between Tokyo and Osaka",
        "originatingSystem": "System_001",
        "priority": 1,
        "reason": "Network infrastructure failure",
        "status": "resolved"
      }
    ],
    "troubleTicket": [
      {
        "@type": "Customer360TroubleTicketVO",
        "href": "https://mycsp.com:8080/tmf-api/troubleTicket/v5/troubleTicket/3180",
        "id": "3180",
        "creationDate": "2019-05-31T07:34:45.968Z",
        "description": "I do not accept the last VOD charge, since the movie was
constantly interrupted, I had to quit watching the movie in the middle",
        "expectedResolutionDate": "2019-06-10T07:34:45.968Z",
        "externalIdentifier": "213-9909",
        "lastUpdate": "2019-06-01T10:15:30.500Z",
        "name": "Compliant over last bill",
        "priority": "High",
        "requestedResolutionDate": "2019-05-31T07:34:45.968Z",
        "severity": "Urgent",
        "status": "pending",
        "statusChangeDate": "2019-06-01T10:15:30.500Z",
        "statusChangeReason": "Need more information from the customer",
        "ticketType": "Bill Dispute"
      }
    ]
  }
]
```


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# API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

## Register listener

**POST /hub**

**Description**

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

**Behavior**

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

**Usage Samples**

Here's an example of a request for registering a listener.

**Request**

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

**Response**

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

## Unregister listener

### DELETE /hub/{id}

### Description

Clears the communication endpoint address that was set by creating the Hub..

### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

### Usage Samples

Here's an example of a request for un-registering a listener.

### Request

```
DELETE /api/hub/42

Accept: application/json
```

### Response

```
204
```

## Publish Event to listener

### POST /client/listener

### Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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### Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

### Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

### Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

### Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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# Acknowledgements

## Release History

| Release Number | Date | Release led by: | Description |
|---|---|---|---|
| Release 5.0 | 07/06/2024 | Uwe Sülter<br>Sergei Lukin<br>Deutsche Telekom | Migration to v5 |
| Release 5.0.1 | 20/09/2024 | Abhilash Prasad<br>Jio Platforms Ltd. | 1. Corrected the discrepancies between the parameter names of entity and sub-entity<br>2. Added LoyaltyAccount in place of LoyaltyBalance sub-entity |
| Release 5.0.2 | 24/02/2025 | Abhilash Prasad<br>Jio Platforms Ltd. | Corrected Examples as part of V4 to V5 Migration Activity |
| Release 5.0.3 | 20/03/2026 | Abhilash Prasad<br>Jio Platforms Ltd. | Enhanced User Guide with comprehensive introduction, detailed use cases, and Customer360-specific polymorphism examples |
| Release 5.0.4 | 27/04/2026 | Abhilash Prasad<br>Jio Platforms Ltd. | 1. Removed Customer360RecommendationVO and Customer360PromotionVO from model and examples<br>2. Updated API Dependencies to include TMF635 Usage Management API<br>3. Refactored Customer360LoyaltyAccountVO to use allOf pattern with Customer360AccountVO<br>4. Aligned entity lists across User Guide and Conformance Profile<br>5. Updated conformance documentation for read-only API characteristics<br>6. Deleted CTK and VS folders (to be generated in build 2) |

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| | | | |
|---|---|---|---|
| Release 5.0.5 | 20/06/2026 | Abhilash Prasad<br>Jio Platforms Ltd. | 1. Fixed API description to accurately reflect supported entities<br>2. Made customer.engagedParty attribute required and added @referredType requirement in PartyRef schema<br>3. Fixed required attributes across all Customer360 VO schemas to align with conformance specifications<br>4. Corrected description typos in schemas<br>5. Removed Usage Records reference from User Guide use cases<br>6. Fixed query parameter examples to use correct nested notation<br>7. Added comprehensive Change Log section documenting differences from v4.0 to v5.0 |
| Release 5.0.6 | 10/07/2026 | Abhilash Prasad<br>Jio Platforms Ltd. | 1. Corrected required attributes across Customer360 VO schemas (ProductOrder, Product, Appointment, CustomerBill, LoyaltyAccount, PartyInteraction) to align with their base TMF schemas<br>2. Updated Change Log note to reference the OpenAPI (Swagger) specification file and conformance profile |

## Change Log from v4.0 to v5.0

This section provides a brief overview of the key differences between TMF717 Customer360 API v4.0 and v5.0.

### Schema and Model Enhancements

- Migrated to TMF v5 common schema patterns including updated base types (Entity,

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EntityRef, Extensible, Addressable)

- Enhanced polymorphism support with stricter `@type` and `@referredType` attribute requirements
- Introduced dedicated Value Object (VO) schemas for all nested entities within Customer360 (e.g., Customer360CustomerVO, Customer360AccountVO, Customer360ProductVO)
- Standardized TimePeriod usage across all temporal attributes with consistent validation rules

**Entity and Attribute Updates**

- Replaced `LoyaltyBalance` with `LoyaltyAccount` entity, including comprehensive account balance details
- Added `PartyInteraction` to capture customer interaction history
- Added `PaymentMethod` for complete payment information visibility
- Corrected attribute naming for consistency with base schemas (e.g., `engagedParty` now mandatory in Customer entity)
- Enhanced nested entity structures with proper inheritance patterns using `allOf`

**API Behavior and Conformance**

- Clarified read-only API characteristics - Customer360 provides consolidated views without modification operations (POST, PATCH, DELETE not supported on Customer360 resource)
- Updated conformance profile with detailed mandatory/optional attribute specifications
- Refined query parameter support to use nested notation (e.g., `customer.name`, `customer.status`)
- Aligned notification events with v5 event patterns

**Documentation and Examples**

- Enhanced User Guide with comprehensive use cases and business scenarios
- Updated all resource and operation examples to reflect v5 schema structures
- Added polymorphism-specific examples for ContactMedium and Characteristic extensions
- Improved API description to accurately reflect supported entities

**Note**: For detailed technical specifications and migration guidance, please refer to the

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OpenAPI (Swagger) specification file and the conformance profile.

## Contributors to Document

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